

Chapter 13

Motivation

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Motivation plays a central role in nearly all aspects of behavior in the workplace. Space constraints preclude an exhaustive review of both classic and contemporary research on work motivation. Fortunately, a number of excellent reviews have been conducted over the past 10 to 15 years that provide extensive review of classic perspectives, as well as providing some treatment of emerging perspectives as of the time of their publication (e.g., Diefendorff & Chandler, 2010; Donovan, 2001; Kanfer, 1990; Lord, Diefendorff, Schmidt, & Hall, 2010; Vancouver & Day, 2005). Likewise, in the preceding edition of this *Handbook*, Mitchell and Daniels (2002) provided a broad introduction to major theories in work motivation. Against this backdrop, our aim in this chapter is to emphasize research conducted since the publication of Mitchell and Daniels's review. In particular, our review focuses primarily on work published within the past 5 to 10 years in major industrial–organizational (I-O) and organizational behavior (OB) journals, as well as relevant research from broader psychological journals. We emphasize current and emerging directions, with classic work discussed briefly as necessary to provide historical context for contemporary issues.

Scholars generally agree that motivation refers to internal forces that underlie the direction, intensity, and persistence of behavior or thought (Kanfer, Chen, & Pritchard, 2008). Direction pertains to *what* an individual is attending to at a given time, intensity represents the *amount* of effort being invested in the activity, and persistence represents *for how long* that activity is the focus of one's attention. We share other scholars' position that motivation reflects processes involved in the allocation of limited resources across the nearly infinite range of possibilities (e.g., Dalal & Hulin, 2008; Kanfer et al., 2008; Pritchard & Ashwood, 2007), a theme we elaborate upon later. Researchers frequently utilize behavioral indicators such as direction, intensity, and persistence as proxies for motivation itself (Ployhart, 2008). Nonetheless, the key question remains: What factors influence these proximal outcomes or indicators of motivation? That question is a primary focus of the theories and empirical studies that form

the basis for this review. It is also important to bear in mind that, although performance is often a key criterion of interest to motivational scholars, one may be highly motivated and thus direct considerable time and effort toward a particular work activity without necessarily achieving a high level of performance (Campbell, McCloy, Oppler, & Sager, 1993).

As prior reviews have very effectively highlighted similarity and divergence among classic theories of motivation, we instead organize our review around constructs and phenomena that have received considerable attention in the recent literature. In particular, given their current prevalence (verging on dominance) in the literature, we use goal-based theories of motivation as an organizing framework around which key theories and constructs are discussed. We begin our review by outlining key elements of the goal-based perspective. We proceed by reviewing research on several categories of constructs and phenomena that apply, extend, or otherwise intersect with the core goal processes, anticipatory constructs including expectancies and self-efficacy, affect, personality, temporal dynamics, and multiple-goal self-regulation. Finally, we close with a brief discussion of potential directions for future research on work motivation. We strive to illustrate the breadth of motivation research's relevance by highlighting research conducted across a range of specific contexts and issues, such as workplace safety, work–life balance, ethical behavior, and others.

Overview of Goals and Goal Processes

Goals are, by far, the most prominent construct in the literature on work motivation. Goals refer to “internal representations of desired states, where states are broadly construed as outcomes, events, or processes” (Austin & Vancouver, 1996, p. 338). The word *goal* often brings to mind conscious, deliberative standards that one is mindfully seeking but, as we shall illustrate throughout our review, the goal concept is considerably broader than this. Indeed, goals exist in many forms and derive from many sources. Some goals are more complex cognitive representations of desired states, but operate at least partially outside our conscious awareness. Some goals are focused on short-term concerns, whereas others are longer-term in nature and may involve representations of the self (Lord et al., 2010). In this section, we provide an overview of fundamental goal processes common to most goal-based perspectives on motivation.

Goal Setting

Scholars have long recognized that motivational processes consist of at least two stages: *goal setting* and *goal striving* (e.g., Lewin, Dembo, Festinger, & Sears, 1944). Goal setting refers to the processes involved in establishing the desired state(s) one is seeking to attain, whereas goal striving refers to the on-line or in-the-moment processes involved in *pursuing* the goals one has set.

As Mitchell and Daniels (2002) noted in the prior edition of this *Handbook*, much of the research on work motivation across the decades has emphasized the goal-setting aspect of motivation, as exemplified by Locke and Latham's (1990, 2002) Goal-Setting Theory (GST). At its core, GST postulates that adoption of difficult, specific goals results in high performance, as compared to less stringent or ill-defined goals (e.g., “do your best”). It is a parsimonious theory that is easily conveyed and readily implemented in the field. GST was inductively derived from scores of empirical studies conducted across myriad contexts, including many well-controlled laboratory and field experiments that rule out many alternative explanations for their effects. Thus, the major tenets of GST have been shown to be applicable across a wide range of contexts, such as individual and group performance (e.g., Locke & Latham, 1990), safety behaviors (e.g., Ludwig & Geller, 1997), well-being, and life

adjustment (Brunstein, Schultheiss, & Grassmann, 1998), among many others. Indeed, numerous narrative reviews and meta-analyses summarize the substantial empirical support for the major tenets of GST (e.g., Locke & Latham, 1990, 2002; Mento, Steel, & Karren, 1987; Tubbs, 1986).

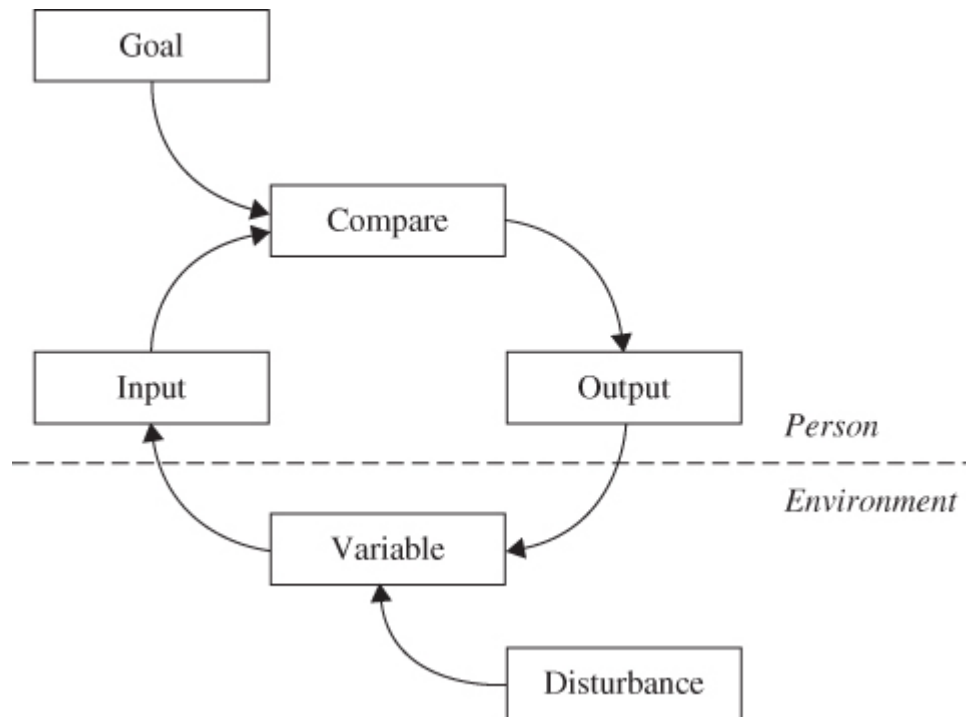
Difficult goals tend to influence performance by fostering greater attention, effort, and persistence, as well as influencing strategy development (Locke, Shaw, Saari, & Latham, 1981; Locke & Latham, 2002). Moreover, numerous moderators have been identified. For instance, difficult goals have more positive effects on performance when individuals are highly committed to their attainment, although goal commitment is less pertinent for easy goals (Klein, Wesson, Hollenbeck, & Alge, 1999). Goals tend to have a stronger positive impact on performance when coupled with feedback that permits individuals to monitor their progress (Locke & Latham, 1990, 2002). Difficult performance goals also tend to have more positive effects on simple or well-learned tasks, and may impair strategy development and performance of novel and/or complex tasks (e.g., Wood, Mento, & Locke, 1987), indicating that difficult performance goals may reliably prompt individuals to “work harder,” whereas their effects on “working smarter” are more complex and potentially detrimental in some circumstances. Additionally, it should be noted that difficult goals may not improve performance if one lacks the ability to perform at the specified level (Locke & Latham, 1990). In such cases, one may lack sufficient confidence in their ability to succeed (i.e., expectancy or self-efficacy) to remain committed to the goal, with disengagement a potential result.

Goal Striving and Self-Regulation

Although goal *striving* has long been of interest to motivational theorists and the subject of some empirical attention, it has traditionally received considerably less attention as compared to goal *setting* (Mitchell & Daniels, 2002). However, one of the more striking trends over the past decade has been an emphasis on the dynamics involved as goals are pursued by people over time, often referred to as self-regulation. In particular, there is a growing body of research rooted in control theory models of self-regulation (e.g., Carver & Scheier, 1998; R. J. Jagacinski & Flach, 2003; Lord & Levy, 1994; Powers, 1973; Vancouver, 2005, 2008).

Discrepancy Reduction

The core of Control Theory models of self-regulation is the discrepancy reduction loop (e.g., Lord et al., 2010), represented in [Figure 13.1](#). The goal level represents the state that the person seeks to attain or maintain. The input function represents the perception of the current state, which is then compared to the desired state. When a gap or discrepancy exists between the current and desired states, the person acts in an attempt to reduce or eliminate the discrepancy. The state of the variable is reassessed, and the process continues. Feedback plays a critical role in that it helps promote alignment between the perceived state and the actual state of the environment. Yet, feedback is often vague, infrequent, delayed, inaccurate, untrusted, or otherwise suboptimal, which can create numerous problems for self-regulation (e.g., Kluger & DeNisi, 1996; Locke & Latham, 1990; Vancouver & Day, 2005).

Figure 13.1 Control loop

Research has found broad support for the influence of discrepancies between current and desired states on subsequent cognitive, affective, and behavioral responses. For example, in a study examining daily fluctuations in effort devoted to job-search activities, Wanberg, Zhu, and Van Hooft (2010) found that lower perceived job-search progress on a given day was associated with greater effort the next day, whereas greater progress was associated with a subsequent reduction in effort. Zoogah (2010) observed that employees may participate in developmental activities as a response to perceived gaps in their competencies, relative to their peers. Yeo and Neal (2008) also provided results consistent with this role of discrepancies, finding that increases in perceived difficulty (i.e., larger discrepancies) were associated with increases in effort, an effect that strengthened with task experience, and was stronger for those with low cognitive ability and low conscientiousness. This link between task difficulty and motivation is consistent with the idea that the resources devoted to a goal are often proportional to the resources needed for success. A number of studies have also demonstrated that discrepancies exert an important influence on shifts in time allocation across multiple goals competing for limited time and attentional resources (e.g., Schmidt & DeShon, 2007). In total, these studies support the notion that the amount of resources devoted to a particular goal can ebb and flow dynamically over time as the goal is pursued, in part as a function of the progress made toward attaining the goal.

Rate of Progress

Velocity or rate of discrepancy reduction may also have important influences on motivational processes (Carver & Scheier, 1998), as does changes in the rate of discrepancy reduction (i.e., acceleration or deceleration) (see Hsee & Abelson, 1991). Lawrence, Carver, and Scheier (2002) provided participants with false feedback indicating that their performance was improving, decreasing, or holding steady across time. Positive velocity resulted in more positive moods, whereas negative velocity resulted in more negative moods. Chang, Johnson, and Lord (2010) assessed perceived and desired job characteristics, as well as individuals' perceived and desired rates of change (i.e., velocity) on those characteristics, finding strong support for the incremental relationship of velocity to job satisfaction, above and beyond the amount of perceived and desired job characteristics. Faster progress was also associated with greater satisfaction and expectations of success, above and beyond the

magnitude of the discrepancies themselves. They further observed that high velocity could compensate for large discrepancies, and small discrepancies could compensate for low velocity; however, the combination of a large discrepancy and low velocity resulted in low expectations for success, reduced satisfaction, and low commitment. Elicker et al. (2010) found a similar relationship of velocity on satisfaction, with this positive relationship strongest when goal importance was also high. Moreover, they observed a positive relationship of velocity on mental focus and goal revision, such that low velocity was associated with reduced attention and lower goals.

External Influences on Goal Progress

It is important to note that the actions of the individual are but one potential source of influence on one's progress toward goal attainment. External influences, referred to as disturbances in control theory parlance, often influence goal progress independent of the action of the individual, moving one closer or further away from the standard. For example, Stewart and Nandkeolyar (2006) found that the number of referrals that salespersons received from their central office had a significant influence on weekly sales performance, contributing above-and-beyond the behaviors directly under the salesperson's control. Such external influences can create highly dynamic situations, and can be an important source of performance variation across time (Stewart & Nandkeolyar, 2006, 2007). A study by Fitzsimons and Finkel (2011) indicates that merely thinking about how other individuals may assist in one's goal pursuits can lead one to reduce the effort expended, presumably due to the perception that fewer personal resources will be required to succeed.

The Intersection of Goal Setting and Goal Striving

Despite important distinctions, the goal-setting and goal-striving phases are often intertwined. This is well illustrated by research on goal revision. This research demonstrates that individuals sometimes respond to discrepancies not by attempting to raise performance up to the standard, but by decreasing goals to better match their performance, a phenomenon often referred to as downward goal revision (e.g., Converse, Steinhauser, & Pathak, 2010; Donovan & Williams, 2003; Elicker, et al., 2010; Ilies & Judge, 2005; Tolli & Schmidt, 2008; Williams, Donovan, & Dodge, 2000). Similarly, although a common response to exceeding one's initial goal is to set a higher performance standard, often referred to as upward goal revision, an alternative response to above-standard performance is to simply maintain or even reduce one's efforts to the original goal (i.e., "coast"). These varying courses of action—that is, whether to persist or disengage when one's performance is deficient, and whether to increase one's ambitions or reduce one's efforts when performance is greater than minimally required for goal attainment—reflect fundamental dilemmas that goal seekers face on a regular basis.

Several studies have shown that attributions can play a role in goal revision, such that goals are revised upward following success and downward following failure to a greater degree if attributed to factors under one's control (e.g., Converse et al., 2010; Donovan & Williams, 2003; Tolli & Schmidt, 2008). Ilies and Judge (2005) and Seo and Ilies (2009) found that affective reactions serve as mediators between performance feedback and subsequent goal setting and performance. Studies have also shown support for Bandura's (1986, 1997) argument that higher self-efficacy is associated with higher goal setting (e.g., Donovan, 2009; Seo & Ilies, 2009; Tolli & Schmidt, 2008). Others have found that self-efficacy *moderated* the effects of discrepancies on goal revision, such that goals were revised upwards following success more readily when accompanied by high self-efficacy (e.g., Converse et al., 2010; Donovan & Hafsteinsson, 2006).

Further illustrating the interplay of goal setting and goal striving is a computational model developed by Scherbaum and Vancouver (2010), demonstrating how negative feedback loops (i.e., discrepancy *reduction*) can result in upward goal revision (i.

e., discrepancy *production*). The crux idea underlying their model is that goals may be increased at one level as a means to reducing discrepancies for a superordinate goal. The output of their model closely matched the behavior of actual participants engaging in a scheduling task, demonstrating that discrepancy-reducing feedback loops can also result in discrepancy production.

Goal Hierarchies and Means–Ends Relationships

Another key element of many models of motivation is that goals are arranged in a means–ends hierarchy, whereby relatively high-level goals (e.g., write a handbook chapter) are attained through the creation and/or activation of relevant subgoals (e.g., write a section), which themselves are accomplished via lower level goals (e.g., review the literature), and so on. Higher level, or superordinate, goals represent the “why” underlying a particular subordinate goal; conversely, the subordinate goal(s) represent the “how” for a particular superordinate goal. There is general agreement that superordinate goals can have several impacts on the functioning of subordinate goals. First, as discussed previously, higher order goals can influence the standards for lower level agents (e.g., Scherbaum & Vancouver, 2010). Second, superordinate goals can influence the strength of the reaction to a given discrepancy between one's actual and desired states (e.g., Hyland, 1988; Jagacinski & Flach, 2003). Reactions to discrepancies are amplified when their success or failure contributes to attainment of highly valued superordinate goals, as compared to discrepancies on goals that are means to less valued ends. Third, superordinate goals can increase the cognitive accessibility of associated means, while also inhibiting the accessibility of competing intentions (e.g., R. E. Johnson, Chang, & Lord, 2006).

One way in which such means–ends relationships have been examined is via monetary outcomes—salary raises, bonuses, incentives, and so forth. Financial inducements frequently result in greater effort and performance, as well as attraction and retention of employees (Jenkins, Mitra, Gupta, & Shaw, 1998; Peterson & Luthans, 2006; Shaw & Gupta, 2007; Stajkovic & Luthans, 2003). Schmidt and DeShon (2007) found individuals were more responsive to goal–performance discrepancies on rewarded than unrewarded tasks, supporting the contention that superordinate goals influence behavior in part due to increasing sensitivity to discrepancies. Interestingly, Shaw, Duffy, Mitra, Lockhart, and Bowler (2003) found that responses to merit pay depend on an individual's level of positive affectivity; because those high on positive affectivity are more sensitive to rewards, they reacted more positively (higher positive affect and intentions to work harder) to merit pay increases than did those low on positive affectivity. Thus, although extrinsic rewards have frequently been found to be effective, their effectiveness is not universal, and many additional motivators remain, some of which are discussed below.

Goal Content

Whereas much of the work motivation literature has focused on general propositions regarding the process by which goals are pursued (sometimes referred to as “structure theories”), content theories of self-regulation instead “describe the types of activities that individuals pursue” and their effects on self-regulation (Diefendorff & Lord, 2008, p. 155). That is, content theories consider *what* is being regulated in terms of the qualitatively different goals. By investigating the structure of goal contents across 15 cultures, Grouzet et al. (2005) developed a two-dimensional goal circumplex model. The first dimension ranges from intrinsic to extrinsic goals, whereas the second dimension ranges from self-transcendence to the physical self. The endpoints of these goal dimensions imply four motivational systems that people navigate regardless of their cultural situation. Intrinsic goals are those that satisfy psychological needs (Gagné & Deci, 2005; Grouzet et al., 2005), a perspective supported largely by Self-Determination Theory (SDT; Ryan & Deci, 2000). In particular, SDT suggests that intrinsic goals satisfy the psychological needs for autonomy, competence, and relatedness, whereas extrinsic goals do not. Extrinsic goals concern social reward and praise (Grouzet et al., 2005), although in the workplace, the most prominent extrinsic goals are likely monetary—

salary raises, bonuses, incentives, and so on. Although research reviewed previously indicates that extrinsic rewards, such as financial incentives, often increase performance (e.g., Jenkins, et al., 1998; Peterson & Luthans, 2006; Shaw et al., 2003; Stajkovic & Luthans, 2003), a review by Gagné and Forest (2008) indicates that pay-for-performance may be most effective for simple and/or boring tasks, and less effective or even potentially detrimental on complex or interesting tasks. Returning to Grouzet et al.'s circumplex, the endpoints of the other goal dimension are self-transcendence and the physical self. Self-transcendent goals refer to universal meanings and understanding, including spirituality, whereas physical goals refer to bodily pleasures and material success. To date, relatively little research has examined self-transcendence and physical goals in the workplace, suggesting an opportunity for future research.

A prominent category of content theories pertain to whether one is seeking to learn and develop, or seeking to maximize performance. These goal types are often referred to as goal orientations (e.g., DeShon & Gillespie, 2005) or achievement motivations (e.g., Cury et al., 2002). Here, we focus on research conducted on the goal content itself, saving examination of individual differences in learning versus performance goals for later discussion. Much of this work has been conducted in a training context. In one such study, Kozlowski et al. (2001) found that emphasizing learning goals led to increased understanding of relationships between task concepts (i.e., knowledge structures), which in turn predicted performance on a more challenging version of the task. Likewise, Seijts, Latham, Tasa, and Latham (2004) found that assignment of difficult, specific learning goals resulted in greater performance than difficult, specific performance goals and “do your best” goals on a complex decision-making task. Kozlowski and Bell (2006) found that distal learning goals (i.e., a focus on learning as much as possible by the end of the training session with little concern about short-term performance), when paired with a mastery goal frame (i.e., viewing performance as malleable and subject to improvement with effort) yielded the most effective self-regulatory behaviors, and thus, the highest end of training performance. Although these studies may imply the general superiority of learning goals over performance goals, this is not necessarily the case. For instance, Chen and Mathieu (2008) found that specific combinations of goals and feedback (namely, learning goals paired with normative feedback and performance goals paired with self-referent feedback) resulted in the greatest rate of improvement on a logic task. Thus, both learning and performance goals are important, as well as the ability to pursue the right goals at the right time; employees and organizations that strike an adequate balance are likely to be most effective (DeShon & Gillespie, 2005).

Goal Framing

In addition to goal content, or *what* is being regulated, there are also goal frames, which refer to how a given goal is construed by or presented to people in a given situation. That is, objectively similar goals may experientially differ depending upon the goal frames. Perhaps the most common goal frame is the contrast between approach and avoidance goals (Carver & Scheier, 1998). Approach and avoid goals play a role in a number of different theories, including theories of achievement goal and goal orientation (e.g., Hulleman, Schragar, Bodmann, & Haraciewicz, 2010; Payne, Youngcourt, & Beaubien, 2007) and regulatory focus theory (Higgins, 1997). In the language of control theories, approach goals are characterized by discrepancy-*reducing* loops, discussed previously, whereas avoid goals are characterized by discrepancy-*increasing* loops (maintaining or increasing the distance from an undesired state). Similar to the approach-versus-avoid distinction, goals can also be framed in terms of gains or losses. Much of this work has roots in Prospect Theory (Kahnemann & Tversky, 1984), which demonstrates that individuals are more sensitive to losses than they are to objectively equivalent gains. Another similar goal frame concerns promotion of positive outcomes versus prevention of negative outcomes (Higgins, 1997). Goals framed in terms of a prevention focus relate to duties and obligations, whereas goals framed in terms of promotion relate to ideal outcomes. Prevention frames lead people to work more slowly, limiting mistakes and “errors of commission,” whereas promotion frames lead people to work more quickly,

limiting “errors of omission” (Förster, Higgins, & Bianco, 2003). People also tend to remain more committed to prevention-focused goals, even when the expectancy of success and the objective value of the associated outcomes are relatively low (Shah & Higgins, 1997). Likewise, people are better able to suppress cognitions related to competing goals when striving to complete a prevention-focused goal as compared to when striving to complete promotion-focused goals (Shah, Friedman, & Kruglanski, 2002). Promotion and prevention frames can be influenced in a variety of ways, such as exposure to prime words related to promotion (e.g., aspiration) or prevention (e.g., duty), or having individuals vicariously experience a promotion versus prevention framed scenario (Friedman & Förster, 2010). In the workplace, a strong safety climate has been shown to predict a prevention focus, leading to safer work behavior (Wallace & Chen, 2006). Likewise, a promotion focus has been shown to mediate a link between leadership behaviors and OCBs (a positive relationship), whereas a prevention focus has been shown to mediate a link between leadership behaviors and CWBs (a negative relationship) (Neubert, Kacmar, Carlson, Chonko, & Roberts, 2008).

Implicit theories of ability are also relevant to goal framing, reflecting whether ability is viewed as relatively malleable (an incremental theory of ability) or relatively unchangeable (an entity theory; Dweck, 2008). When abilities are framed as malleable, people believe that skills can be developed via effort and practice, and thus tend to be more resilient in the face of difficulties and set-backs. On the other hand, when abilities are framed as innate and largely static, low performance is seen as evidence of a lack of ability, often resulting in withdrawal from the task. Data from the classroom (e.g., Blackwell, Trzesniewski, & Dweck, 2007) and from the training literature (e.g., Kozlowski & Bell, 2006) both suggest that the encouragement of an incremental theory (i.e., skills are malleable), as compared to encouragement of an entity theory (i.e., skills are fixed), results in better learning and performance outcomes.

Nonconscious Self-Regulatory Processes

Although the discussions thus far may seem to imply a conscious, deliberative process whereby people explicitly take stock of their progress repeatedly as a goal is pursued, this need not be the case. Theory and research converge on the notion that such monitoring can, and often does, operate outside of conscious awareness (e.g., DeShon & Gillespie, 2005; R. E. Johnson et al., 2006; Wegner, 1994). Indeed, Lord and Levy (1994) argued that effective self-regulation, including adaptation to changing environmental conditions, requires parallel monitoring of many goals simultaneously, such that attention can be redirected from one's current concerns toward other pressing matters. If discrepancy monitoring could occur only through conscious deliberation, such massively parallel monitoring would quickly overburden attentional resources, whereas monitoring only what can be consciously reflected upon would leave individuals incapable of complex, adaptive behavior. Responses to detected discrepancies may also be undertaken without conscious reflection, although discrepancies are capable of capturing and redirecting conscious attention when necessary.

Goal Activation

For an implicit goal to influence behavior, it must first be activated. Goal activation may occur through the mechanisms of priming and spreading activation. Priming occurs when exposure to stimuli in one's environment activates a nonconscious goal (Bargh & Chartrand, 2000). For example, Holland, Hendriks, and Aarts (2005) demonstrated that participants who were exposed to the scent of all-purpose cleaner left fewer crumbs after eating a cookie than participants who were not exposed to this scent. Importantly, participants in the scent condition did not report being aware of the scent, therefore indicating that the priming happened below conscious awareness. The authors inferred that the scent of the cleaner activated nonconscious “cleanliness”

goals for participants in the experimental group, leading to the difference in cleanliness behavior across conditions. Stajkovic, Locke, and Blair (2006) used a supraliminal prime (i.e., above conscious awareness) to influence performance on a creativity task. Participants in the experimental condition solved word-search puzzles (pilot study) or unscrambled sentences (main study) consisting of achievement-related words (e.g., *succeed*, *strive*, *attain*), while participants in the control conditions completed similar puzzles consisting of neutral words (e.g., *turtle*, *green*, *lamp*). The authors found that primed participants were able to generate significantly more uses for common household items (wire coat hanger, wooden ruler), which is a common creativity task. Furthermore, priming predicted performance incrementally beyond consciously assigned goals (e.g., “Generate 12 uses”), and the effects of the non-conscious goals persisted up to one day later. More recently, Shantz and Latham (2009) demonstrated the usefulness of nonconscious goals in a field setting, showing that call center workers exposed to an achievement prime (a picture of a woman winning a race) generated more money in donations than workers who were not exposed to the prime.

Goal activation also occurs through a process known as spreading activation; when a goal is activated, related goals and knowledge are also likely to be activated (e.g., Lord and Levy, 1994). A meta-analysis by R. E. Johnson et al. (2006) found activation of a goal increases the speed at which information pertaining to the goal is retrieved from memory, as well as the likelihood that such information will be retrieved. For example, priming a goal of grocery shopping has been shown to lead to increased activation of means of getting to the store, such as the bus or a bicycle (Aarts & Dijksterhuis, 2000). Likewise, automatic activation of a goal (e.g., studying) can lead to more favorable judgments of stimuli that are useful for pursuing the goal (e.g., a library) (Ferguson, 2008). Individuals are also more likely to attend to stimuli in their environment that are useful for pursuing an active goal than stimuli that are not useful for goal pursuit (e.g., Vogt, De Houwer, Moors, Van Damme, & Crombez, 2010). Goal activation does not only spread from higher order goals to means of achieving them; activation of means goals can also activate the higher order goals they serve (Shah & Kruglanski, 2003).

Goal Inhibition

While the activation of a goal can lead to the *activation of related goals*, it can also lead to the *inhibition of competing goals*. That is, when a goal is activated, alternative goals that compete for the same resources are actually suppressed, making them less accessible in working memory (R. E. Johnson et al., 2006). Shah et al. (2002) identified a number of conditions under which such goal shielding effects are likely to emerge. These authors found that goal commitment is positively related to inhibition of alternative goals. Alternative goals that *facilitate* focal goal pursuit are less likely to be shielded than goals that *compete* with the focal goal, and goals construed as *duties or obligations* are more likely to be shielded than goals construed as *ideals*. However, when individuals must switch between goals (e.g., when interruptions occur), individuals may find it difficult to inhibit cognitions related to the initial task while they are performing the interrupting task—a phenomenon termed “attention residue” (Leroy, 2009). Leroy (2009, 2010) showed that, when individuals switch from one task to another before the first task is completed, they are often unable to inhibit cognitions related to the first task. This effect is more likely when individuals perceive insufficient time to complete the initial task, as anxiety about failure leads cognitions regarding the first task to persist. Leroy shows that attention residue is *negatively* related to task performance on the interrupting task, just as goal shielding is *positively* related to task performance (Shah et al., 2002).

Deactivation

Finally, R. E. Johnson et al.'s (2006) meta-analysis demonstrated that goals tend to become less activated—meaning that individuals take longer to respond to goal-relevant stimuli and are less likely to recall goal-relevant information—when goals have been accomplished. Goals may also decrease in activation as individuals realize they will not be able to accomplish them.

For instance, Förster, Liberman, & Higgins (2005) found that information relevant to goals with a high probability of success (90%) was easily accessible to participants, yet information related to goals with little chance of success (5%) was not. However, activation can sometimes persist even after a goal has been accomplished, such as when individuals experience attention residue even after the task is completed (Leroy, 2009).

Expectancies, Self-Efficacy, and Related Constructs

Two related constructs that have received considerable attention throughout the years are expectancy and self-efficacy. Both constructs are prospective and forward looking, pertaining to projections about future performances and, as such, have considerable influence on self-regulation. We begin with expectancy, which has a longer history within the motivation literature. We then focus on self-efficacy, which has been the more dominant prospective construct within the I-O/OB literature over the past two decades. Finally, we close this section by discussing some other relatively new constructs that bear some similarities to, as well as important distinctions from, expectancy and self-efficacy.

Expectancies

Expectancy has been a focal construct at least since the 1930s (e.g., Lewin, 1935). Broadly, expectancy refers to the perceived likelihood that an action will lead to a particular outcome. It has traditionally been paired with the concept of valence, which refers to the attractiveness of an outcome, to form the core of expectancy-value theories (e.g., L. Porter & Lawler, 1968; Vroom, 1964)—often simply referred to as Expectancy Theory. These theories propose that expectancies and valence jointly determine the tendency to act in a particular way, often referred to as *utility* or *motivational force*, with actions possessing greater expectancy and value being more likely to be exhibited. Vroom's (1964) variation of Expectancy Theory gained a particular foothold in the work-motivation literature. An important feature of Vroom's theory is the addition of Instrumentality. Whereas expectancy within Vroom's theory refers more specifically to the subjective probability that a given level of effort will result in a given level of performance, instrumentality refers to the perceived likelihood that a given level of performance will result in secondary outcomes such as pay, recognition, and so on. Vroom proposed that valence, instrumentality, and expectancy combine to influence choice.

Expectancy theory has been utilized to understand and predict various aspects of motivated behavior, such as choice among self-set goal levels, acceptance of and commitment to assigned goals, among other outcomes (Kanfer, 1990). Although limited support has been obtained from between-person analyses of expectancy-based motivation theories—for example, individuals with higher subjective utility (i.e., expectancy $\times$ valence) for a particular goal do not necessarily outperform individuals with lower motivational force for that same goal—the theory was originally proposed as a *within-person* theory of how individuals choose among alternatives. Tested in this manner, the results have been more compelling (Van Eerde & Theirry, 1996). For example, individuals presented with a set of potential goal difficulty levels are more likely to select the goal level with the greatest subjective utility (e.g., Klein, 1991).

Researchers continue to utilize the expectancy concept, although perhaps less frequently and directly than in the past. For example, in an application of expectancy theory to applicant self-selection, Kuncel and Klieger (2007) reasoned that low

expectations for successful admission to highly prestigious law schools underlies the large disparity in LSAT scores observed between applicants to highly prestigious law programs and applicants to less prestigious law programs. Reinhard and Dickhäuser (2009) demonstrated that expectancies positively related to performance on difficult tasks, but only if difficulty is taken into consideration when the expectancy is formed. Ames (2008) showed that expectancies regarding effectiveness of assertive behavior predicts the amount of assertiveness exhibited in the workplace. Together, these results further demonstrate that expectancies often foster willingness to undertake difficult endeavors. Researchers have also identified antecedents of expectancies. It has long been established that, all else being equal, expectancies tend to decrease as difficulty increases (Locke & Latham, 1990). In a similar vein, goal progress and time jointly influence expectancies, as large discrepancies present a greater challenge when little time remains to resolve them and/or when velocity is low (e.g., Schmidt & Dolis, 2009; Chang et al., 2010). However, Dickhäuser and Reinhard (2006) found that individuals sometimes fail to recognize the true difficulty of a task, with unduly high expectancies as a result. This was particularly the case when cognitive capacity and/or need for cognition were low, both of which tended to discourage sufficient reflection on the difficulty of the task.

In our view, some of the most intriguing research concerning expectancies pertains to their role in multiple-goal self-regulation; that is, in the choice of which of multiple competing demands one chooses to pursue from one moment to the next. Steel and König's (2006) Temporal Motivation Theory, which integrates expectancy-value theories with other related perspectives, holds substantial promise in this regard, and is detailed later in this review. In many respects, this research is getting back to the etiological roots of expectancy theories, as inherently within-person models of the processes by which one chooses from among a set of alternatives. We discuss these issues in greater detail in a later section of this review, focused explicitly on multiple goal research. For now, we turn our attention to the more commonly utilized variation of expectancy notions among contemporary motivation scholars: self-efficacy.

Self-Efficacy

Self-efficacy has been among the most widely studied constructs in motivation. Self-efficacy is defined as “beliefs in one's capabilities to mobilize the motivation, cognitive resources, and courses of action needed to meet given situational demands” (Wood & Bandura, 1989, p. 408). Self-efficacy is generally viewed as a positive contributor to a wide range of beneficial processes and outcomes across an even broader range of contexts, including the workplace, academics, athletics, and many others (e.g., Moritz, Feltz, Fahbrach, & Mack, 2000; Multon, Brown, & Lent, 1991). Self-efficacy influences performance via a variety of mediating mechanisms, such as setting challenging goals, allocating time, effort, and other resources to those goals, and persisting with one's goals in the face of adversity (Bandura, 1997). Of particular interest within the work-motivation domain, considerable research has demonstrated a positive relationship between self-efficacy and task performance. Meta-analyses by Stajkovic and Luthans (1998) and by Judge, Jackson, Shaw, Scott, and Rich (2007) found mean correlations of 0.38 and 0.37, respectively, between self-efficacy and performance. However, as we shall soon discuss, there has been some debate over the past decade concerning the magnitude and even direction of self-efficacy's effects on performance.

Beyond task performance, researchers have also demonstrated positive relationships between self-efficacy and myriad outcomes, such as commitment to organizational change (e.g., Herold, Fedor, & Caldwell, 2007), entrepreneurship (e.g., Hao, Seibert, & Hills, 2005), and creativity (e.g., Gong, Huang, & Farh, 2009), among many others. Self-efficacy has also been implicated as a key process variable involved in leadership. For example, transformational leadership has been found to increase employee creativity in part by increasing employee self-efficacy (Gong et al., 2009; Walumbwa, Avolio, & Zhu, 2008). Additionally, empowering leadership behaviors, such as providing autonomy and fostering participative decision making, are associated with higher subordinate self-efficacy (Ahearne, Mathieu, & Rapp, 2005).

Despite the large body of research indicating the beneficial effects of self-efficacy, a contentious issue among motivational scholars has been whether self-efficacy truly exerts a positive effect on subsequent performance, or whether the observed positive relationships are spurious. Vancouver and colleagues (Vancouver & Kendall, 2006; Vancouver, Thompson, Tischner, & Putka, 2002; Vancouver, Thompson, & Williams, 2001) argued that the positive relationship of self-efficacy with performance may be a spurious result of past performance's strong influence on subsequent self-efficacy; however, they went further to suggest that the effect of self-efficacy on subsequent performance may actually be negative, albeit modest in magnitude, resulting from higher self-efficacy facilitating a belief that fewer resources are needed to attain the goal in question. They further argued that, due to the strong positive effect of past performance on subsequent self-efficacy, cross-sectional research designs might mask any negative effects that may exist. In several studies utilizing longitudinal designs, whereby self-efficacy and performance were tracked across multiple trials, support was found for these arguments (Vancouver & Kendall, 2006; Vancouver et al., 2001, 2002). Subsequent research has replicated the null or negative relationship of self-efficacy and performance at the within-person level of analysis, in both the lab (e.g., Heggstad & Kanfer, 2005; Richard, Diefendorff, & Martin, 2006, Study 2; Yeo & Neal, 2006) and the field (e.g., Richard et al., 2006, Study 1; Vancouver & Kendall, 2006; Wanberg et al., 2010). A meta-analysis by Judge et al. (2007) concluded that the positive relationship between self-efficacy and performance largely disappears after accounting for personality (Big Five), experience, and cognitive ability.

Yet, as Bandura and Locke (2003) note, many prior studies utilizing within-person methodology, including within-person experimental manipulations, have observed positive relationships of self-efficacy with effort and performance. In an effort to understand such variability, research by Schmidt and DeShon (2009, 2010) has sought to identify moderators of the self-efficacy's effects. Schmidt and DeShon (2010) examined the moderating role of performance ambiguity, finding a negative relationship when individuals' true performance was highly ambiguous. With ambiguity, individuals are thought to draw upon their self-efficacy perceptions to estimate their performance, with positively biased estimates leading individuals to invest less time and effort than is required. However, for participants whose true performance was unambiguous, the potentially biasing effects of self-efficacy on performance perceptions was inhibited, mitigating the potential for a negative effect. Schmidt and DeShon (2009) found a positive relationship when individuals faced a challenging situation, a relationship attributed to high self-efficacy promoting persistence in the face of adversity (e.g., Bandura, 1997). In contrast, when confronting lesser challenges, a negative relationship was observed, which was ascribed to high self-efficacy fostering a belief that continued success could be attained with minimal effort. Similarly, Schmidt and Beck (2011) found a positive relationship among those assigned a difficult goal, but a negative relationship with an easy goal. Beck and Schmidt (2011) demonstrated that the effects of an increase or decrease in self-efficacy differ for individuals who are already highly efficacious compared to those whose efficacy is more modest.

Although these studies indicate the potential for self-efficacy to impair performance, these processes are thought to reflect generally adaptive, beneficial functions (Vancouver, 2005). That is, self-efficacy is utilized in an attempt to allocate one's limited resources as efficiently as possible. High self-efficacy often suggests fewer resources (e.g., time, energy) need to be allocated to the task at hand, thus allowing resources to be conserved for other purposes. Indeed, Vancouver, More, and Yoder (2008) found higher self-efficacy was associated with less time allocated to a given task trial, saving time for subsequent trials that could be more difficult and, thus, in need of additional time. In a test-taking context, Beck and Schmidt (2011) observed a negative relationship between self-efficacy and time allocated to a block of test items among those given limited time to complete the test. This allowed more items to be completed within the available time and/or saved limited time for difficult questions that could subsequently appear. In contrast, a positive relationship between self-efficacy and time allocation was found among test takers given unlimited time to complete the test, likely due to higher self-efficacy leading individuals to view more effort as facilitating greater performance, thus justifying the use of more time.

Other Self-Efficacy/Expectancy-Related Constructs

Whereas self-efficacy is generally regarded as a task-specific belief, researchers have also examined *general* self-efficacy (GSE), which is a more global perception of individuals' perceived capability to succeed in a broad range of tasks and situations (Chen, Gully, & Eden, 2001). Like its task-specific counterpart, individual differences in GSE have often been found to relate positively to performance, although those effects appear to be mediated at least in part by task-specific self-efficacy (e.g., Chen, Gully, Whiteman, & Kilcullen, 2000; Yeo & Neal, 2008). Even broader than GSE is Core Self-Evaluations (CSE), which refers to one's global assessments of their worth and competence (e.g., Judge, Bono, Erez, & Locke, 2005). We discuss CSE further in a later section on individual differences.

Eden (2001) proposes that individuals' perceptions of their own capabilities is only part of the story, and that one's assessments of task-relevant external resources—external efficacy—may have an important influence on motivation and performance. With a pessimistic view of the external resources, expectancies for success, and thus effort, may be low even if one has high self-efficacy. One form of external efficacy that has received some attention is means efficacy, defined as one's belief in the usefulness of the tools available for performing the job. For example, Eden, Ganzach, Flumin-Granat, and Zigman (2010) found that participants informed they would be using a state-of-the-art computer system had higher means efficacy and performance than control participants, despite the controls utilizing the same computer system.

Collective efficacy goes beyond individual self-efficacy to reflect perceptions of a *group's* capabilities to succeed on the task at hand. Like self-efficacy, a meta-analysis of collective efficacy shows that it typically relates positively to team performance (Stajkovic, Lee, & Nyberg, 2009), mediating the effects of various factors such as shared mental models (Mathieu, Rapp, Maynard, & Mangos, 2010) and empowering team leadership (Srivastava, Bartol, & Locke, 2006). Teams with high collective efficacy are also more likely to successfully adapt their training to more complex and challenging environments (Chen, Thomas, & Wallace, 2005). Collective efficacy has also been found to exhibit cross-level influences on individual performance within a team context, as individuals possess greater self-efficacy and set higher individual goals for themselves when they are part of a team they believe to be highly capable (Chen, Kanfer, DeShon, Mathieu, & Kozlowski, 2009). However, as is the case with self-efficacy, there are also indications that collective efficacy may have its downsides as well. For example, Goncalo, Polman, and Maslach (2010) found that teams that became highly confident too early in the groups' existence were less likely to engage in beneficial conflict and debate regarding how the team should undertake its task, which in turn contributed to lower performance among these teams. In contrast, they found that high collective efficacy and low process conflict were beneficial when they occurred later in the teams' development.

Affect

Affect and motivation often go hand-in-hand. *Affect* is an “umbrella term encompassing a broad range of feelings that individuals experience, including states, such as moods and discrete emotions, and traits, such as trait positive and negative affectivity” (Barsade & Gibson, 2007, p. 38). Affective experiences include *emotion*, which is directed at someone or something, and *mood*, which is less intense, longer lasting, and not directed at a specific target (Lord & Kanfer, 2002). Historically, the most prominent approach to studying affect in the workplace is *job satisfaction* (Brief & Weiss, 2002), generally defined as one's feelings about the job situation (Smith, Kendall, & Hulin, 1969). Many scholars suggest job satisfaction has both affective and cognitive

components and thus is distinct from emotion (e.g., Weiss & Beal, 2005). Emotions are theorized to influence job attitudes as a result of affect-inducing events (Weiss & Cropanzano, 1996). Thus, emotions experienced at work are more likely an antecedent of job satisfaction. Still, in applied settings, job satisfaction remains a popular way to study affect in the workplace; thus we integrate work on job satisfaction into our review where appropriate. We review (a) affect as an antecedent of motivation and behavior, (b) affect as an outcome of motivation and behavior, and (c) the self-regulation of emotions in the workplace.

Affect as an Antecedent of Motivation and Behavior

In addition to being a meaningful outcome in its own right (Weiss & Rupp, 2011), affect matters in part because it influences various outcomes of concern in the workplace. There is a long-standing interest in the hypothesis that “a happy worker is a productive worker” (see Kluger & Tikochinsky, 2001). In general support of this hypothesis, both positive and negative affect (Kaplan, Bradley, Luchman, & Haynes, 2009) and job satisfaction (Iaffaldano & Muchinsky, 1985; Judge, Thoresen, Bono, & Patton, 2001; Riketta, 2008) have been shown to correlate with job performance. Similarly, Miner and Glomb (2010) found that periods of positive mood were associated with periods of improved performance. The authors drew upon theory to suggest that mood precedes behavior, but acknowledged that reciprocal causation may also be occurring.

The effects of affect on performance may be driven by persistence, such that individuals are more likely to “stick with” a task they enjoy. Specifically, there is longitudinal research to suggest a link between positive moods and task performance that is in part mediated by the motivational processes of self-efficacy and task persistence (Tsai, Chen, & Liu, 2007). Also, in a meta-analysis by Kaplan et al. (2009), the authors found motivational process variables to partially mediate relationships between positive and negative affectivity and task performance. For example, Seo and Ilies (2009) found that individuals set higher goals, spent more time pursuing those goals, and achieved higher levels of performance when experiencing more positive affect. Seo and Ilies also found that the effects of affect on goal setting were mediated by self-efficacy. Likewise, Erez and Isen (2002) showed that positive affect enhanced perceptions of valence, instrumentality, and expectancy for a task.

Aside from job and/or task performance, affect has been linked to a variety of other important organizational outcomes as well. For instance, job satisfaction relates negatively to turnover (e.g., Griffeth, Hom, & Gaertner, 2000), and *decreases* in job satisfaction over time may be even more predictive of turnover than absolute levels of job performance, as employees may “believe that their experience at work will ‘stay the course’ (i.e., sustain a downward trend)” (Chen, Ployhart, Thomas, Anderson, & Bliese, 2011, p. 176). Also, employees may be more inclined to exhibit organizational citizenship behaviors when they are experiencing positive affect, yet they may exhibit more counterproductive work behaviors when experiencing negative affect (Dalal, Lam, Weiss, Welch, & Hulin, 2009). However, in a similar study, Conway, Rogelberg, and Pitts (2009) showed that positive affect predicted helping behavior only for those individuals relatively high in the personality trait of altruism. Finally, affect is linked to decision-making behavior. For instance, Seo, Goldfarb, and Barrett (2010) found affect to mitigate the role of decision frames (i.e., gains and losses) in subsequent risk-taking. The authors found that the tendency to avoid risk after experiencing gains disappeared or even reversed when people simultaneously experienced pleasant feelings. This finding is in line with Friedman and Förster’s (2010) contention that positive affect signals a “benign situation,” meaning that individuals can afford to be less cautious and more apt to explore opportunities.

Affect as an Outcome of Motivation and Behavior

Affect is also an outcome of goal progress. Success tends to be associated with positive affect, whereas failure or slow progress tends to be associated with negative affect (e.g., Carver & Scheier, 1998). In a relatively direct test of the affective consequences

of goal progress, Chang et al. (2010) found task satisfaction and task motivation during goal-striving depends not only on goal discrepancies but also on *velocity*, or the rate at which goal discrepancies change over time. Further, several studies have demonstrated the applied implications of the effect of goal progress on affect. For instance, Wanberg et al. (2010) showed that daily perceived progress in a job search process was related in the expected direction to vacillation in the job seekers' positive and negative affect. Finally, Rogelberg, Leach, Warr, and Burnfield (2006) showed a link between perceived meeting effectiveness and job-related affect (i.e., comfort, enthusiasm).

Although we have focused on goal progress influencing affect and affect influencing behavior (and thus, goal progress), it should be clear that the interplay between goals and affect is likely reciprocal. For example, Ilies and Judge (2005) manipulated perceptions of goal progress via positive and negative feedback. These authors found that feedback predicted momentary affect, which in turn predicted the subsequent goals individuals set. Similarly, Cron, Slocum, VandeWalle, & Fu (2005) provided negative feedback to participants, showing a link between negative emotional reactions to the feedback and subsequent goal level individuals set for themselves. However, the effect of negative emotions on goal setting depended on goal orientation, such that negative emotional reactions led to decrements in subsequent goal level only for individuals relatively low in learning orientation. Those with high learning orientation were more resilient, keeping their goals high even when experiencing negative emotions.

Emotional Labor and Emotion Regulation

Individuals frequently seek to regulate their emotional experiences and/or emotional displays in the workplace. *Emotion regulation* refers to the psychological processes “influencing which emotions one has, when one has them, and how one experiences and expresses these emotions” (Gross, 1998). The self-regulation of emotion may be used for a wide range of purposes, such as coping with stressful situations (Lazarus, 1975, 1999; Lazarus & Folkman, 1984). Emotion regulation includes not only the control of overt behavior associated with an emotion (e.g., expressive gestures) but also “the entire organized state that is subsumed under the emotion construct” (Lazarus, 1975, p. 57). *Emotional labor* is similar in many regards to emotion regulation, but refers to employees' management of their feelings, or their apparent feelings as viewed by customers and coworkers, in accordance with organizationally defined rules and guidelines (Wharton, 2009). Grandey (2000) incorporated existing research on emotional labor and emotion regulation, along with affective events theory (Weiss & Cropanzano, 1996), to present a comprehensive model of emotional labor and its antecedents and consequences. Her model distinguishes between *antecedent-focused* and *response-focused* emotional labor strategies (e.g., Grandey, 2003). Antecedent focused strategies, such as deep acting (“faking in good faith”), refer to changing one's internal states to match organizational expectations. In contrast, *response-focused* strategies, such as surface acting (“faking in bad faith”), involve changing only one's external displays to *appear* as if one is experiencing the expected emotions.

Both strategies appear to have advantages and disadvantages. When affective shocks at work (e.g., interpersonal conflict) elicit negative emotions (see Grandey & Brauburger, 2002), it may be quicker and easier to deploy a response- (vs. antecedent-) focused strategy, as there may not be sufficient time to modify internally felt emotions, particularly when these events are unexpected (Diefendorff & Gosserand, 2003). However, surface acting may come at a cost, being associated with lower job attitudes (Côté & Morgan, 2002), greater emotional exhaustion (Grandey, 2003), reduced attentional resources (Goldberg & Grandey, 2007), depersonalization (Brotheridge & Grandey, 2002), and turnover (Chau, Dahling, Levy, & Diefendorff, 2009). Suppressing or concealing of emotion is another example of response-focused emotional labor that has been shown to drain task resources (Wallace, Edwards, Shull, & Finch, 2009) and to be associated with negative job attitudes (Gillespie, Barger, Yugo, Conley, & Ritter, in press). Interestingly, Grandey, Fisk, and Steiner (2005) found employees reporting relatively high autonomy

did not experience emotional exhaustion following surface acting (see also H. M. Johnson & Spector, 2007). Similarly, Trougakos, Beal, Green, and Weiss (2008) found that rest or break activities may mitigate the costs and accentuate the benefits of emotional labor (see also Beal, Weiss, Barros, & MacDermid, 2005). In contrast to surface acting, deep acting may have beneficial effects for employees, such as a greater sense of personal accomplishment (e.g., Brotheridge & Grandey, 2002). Other forms of antecedent-focused emotional labor (e.g., situation selection, situation modification; see Grandey, 2000; Gross, 1998) have not been as well studied in the literature, and thus this represents a direction for future research.

Although much of the motivation literature focuses on goals related to thought or behavior, there is an increasing amount of attention being paid to emotional goals. In a clear example of this, Diefendorff and Gosserand (2003) drew from control theory models of self-regulation (e.g., Carver & Scheier, 1998) along with other theories of motivation (e.g., Locke & Latham, 1990; Ryan & Deci, 2000) to present a theory of why some people are more motivated than others to comply with a given display rule, such as providing service with a smile. In an empirical study, Gosserand and Diefendorff (2005) identified display rule commitment as a moderator of the links between display rule perceptions and outcome variables (namely, surface acting, deep acting, and positive affective delivery at work), with these associations being stronger for those relatively high in commitment. This finding and the many testable propositions contained in Diefendorff and Gosserand's theory suggest that display rules are a goal or standard toward which people strive and that the motivational processes underlying these pursuits may be largely similar to those underlying task goals.

Individual Differences Related to the Self and Personality

There is a long history in psychology of interest in the self (see Baumeister, 1998), psychological needs (Kanfer, 1990; Sheldon, Elliot, Kim, & Kasser, 2001), and personality (McAdams & Olson, 2011). Although there is some debate, scholars generally agree that the self is important in that it interprets and organizes relevant actions and experiences (Markus & Wurf, 1987). Needs may be defined as particular qualities of experience that all people require to thrive (Deci & Ryan, 2000). Finally, personality may be defined as “the dynamic organization within the individual of those psychosocial systems that determine his characteristic behavior and thought” (Allport, 1961, p. 28) and thus tends to convey a sense of consistency, internal causality, and personal distinctiveness (Carver & Scheier, 2008). People may exhibit individual differences in the self and personality in that they may place a greater importance on certain needs or goals and thus exhibit different traits as compared to others.

The Self and Psychological Needs

Self-Determination Theory (Deci & Ryan, 2000; Ryan & Deci, 2000) argues that three basic psychological needs—needs for competence, relatedness, and autonomy—are fundamental to health and well-being, and are the source of intrinsic motivation. *Competence* refers to the need to “have an effect on the environment as well as to attain valued outcomes” (Deci & Ryan, 2000, p. 231). *Relatedness* refers to the desire to feel a sense of attachment and connection with others, and *autonomy* refers to the desire to “self-organize experiences and behavior and to have activity be concordant with one's integrated sense of self” (Deci & Ryan, 2000, p. 231). Based on SDT, only the behavior associated with the pursuit of these intrinsic or integrated goals is “self-determined” (Deci & Ryan, 2000).

Similarly, Motivated Action Theory (MAT; DeShon & Gillespie, 2005) proposes “self-goals” as the highest-order, fundamental goals that everyone strives to achieve, at least to some extent, to lead a healthy and fulfilling life. First, MAT proposes that people strive to achieve and maintain the perception that they can intentionally influence important aspects of the environment, called *agency* (Bandura, 2006). MAT also proposes a desire to achieve and maintain a positive self-image, called *esteem* (Allport, 1955), and a need to form and maintain positive interpersonal relationships with others, called *affiliation* (Baumeister & Leary, 1995). MAT proposes optimal health and well-being arise when there is relatively little discrepancy on self-goals, or at least a perception of adequate progress toward discrepancy reduction. MAT emphasizes how goals lower in the hierarchy, most specifically achievement goals, are set to facilitate pursuit of higher order goals that are relevant to the self. That is, to meet higher order goals of autonomy, esteem, and relatedness people strive to achieve performance outcomes by learning new skills, demonstrating current skills to others, and avoiding demonstrating a lack of skill. Whereas self-goals are pursued over long periods of time, even one's entire life, achievement goals are pursued over shorter time frames (Lord et al., 2010).

There is a growing body of research on the benefits of congruence or fit between psychological needs and the demands of the situation. In a feedback intervention study, Anseel, Lievens, and Schollaert (2009) found that reflection strategies were less effective for individuals low in need for cognition, as they were less likely to engage in reflection after feedback. Greguras and Diefendorff (2009) suggest that the satisfaction of psychological needs (i.e., competence, relatedness, and autonomy) partially explains the relationship between perceptions of person–environment fit and affective commitment and performance. Studies such as these suggest that individuals with higher levels of certain psychological needs may be more or less well suited for certain types of organizational interventions and work situations.

Core Self-Evaluations

A growing body of literature concerns the relatively recently proposed Core Self-Evaluations (CSEs). CSE is defined as “fundamental assessments that people make about their worthiness, competence, and capabilities” (Judge et al., 2005, p. 257). It is a latent construct comprised of the overlapping variance among other, more specific, individual difference constructs such as self-esteem, generalized self-efficacy, locus of control, and emotional stability (Judge, Martocchio, & Thorensen, 1997). CSE has been linked to numerous positive outcomes, such as job performance (Kacmar, Collins, Harris, & Judge, 2009), coping processes (Kammeyer-Mueller, Judge, & Scott, 2009), financial well-being (Judge, Hurst, & Simon, 2009), job-search intensity (Wanberg, Glomb, Song, & Sorenson, 2005), and others. Judge et al. (2005) found that CSE was positively related to job and life satisfaction, in part due to the pursuit of goals that are consistent with one's values (i.e., goal self-concordance). Kacmar et al. (2009) demonstrated that the positive relationship between CSE and job performance was stronger for those who perceived a favorable work environment (i.e., low perceptions of politics and high perceived leadership), suggesting that favorable environments enable the benefits of CSE to manifest. However, other scholars have raised a number of theoretical concerns, including the need for clearer specification and evaluation of the nature of the CSE construct, the mechanisms by which CSE has its effects, how it develops, and the criteria for determining which traits are fundamental enough for inclusion as part of the CSE construct (R. E. Johnson, Rosen, & Levy, 2008; see also Ferris, Lian, Brown, Pang, & Keeping, 2011).

Five Factor Model (FFM)

The FFM is a prominent model of personality—consisting of agreeableness, conscientiousness, extraversion, neuroticism (or emotional stability), and openness to experience—that has been found to be robust across cultures (McCrae & Terracciano, 2005; Yamagata et al., 2006). Although there is research on other traits (e.g., neuroticism; Smillie, Yeo, Furnham, & Jackson,

2006), in recent years, there has been a growing amount of research on conscientiousness and its narrower traits of achievement and dependability (Dudley, Orvis, Lebiecki, & Cortina, 2006; Perry, Hunter, Witt, & Harris, 2010). Conscientiousness has been consistently shown to be a valid predictor across performance measures in all occupations studied (Barrick & Mount, 1991; Barrick, Mount, & Judge, 2001). Further, moderators of the link between conscientiousness and performance have been identified. Colbert and Witt (2009) found that conscientiousness was more strongly positively related to performance among workers who perceived their supervisors to be relatively high in goal-focused leadership. Cianci, Klein, and Seijts (2010) found a focus on performance (vs. learning) goal following negative feedback led to more tension (e.g., feeling jittery, fearful, etc.) and lower performance for those individuals relatively high in conscientiousness, as compared to individuals with lower conscientiousness.

FFM investigations have also extended beyond personality-performance correlations. Self-monitoring has been identified as a moderator (Barrick, Parks, & Mount, 2005), such that there were attenuated relationships between three FFM traits (Extraversion, Emotional Stability, and Openness to Experience) and supervisory ratings of interpersonal performance when individuals were relatively high in self-monitoring. FFM traits also predict other variables of interest besides performance, such as counterproductive work behaviors (Mount, Ilies, & Johnson, 2006) and job search behavior (Turban, Stevens, & Lee, 2009), and there is an increasing number of FFM studies being conducted in unique contexts, such as leadership (Hendricks & Payne, 2007; Ng, Ang, & Chan, 2008). Moreover, the FFM has been integrated into other theories and bodies of research, such as that on trait-consistent affect (Bono & Vey, 2007; Tamir, 2005) and organizational justice (Colquitt, Scott, Judge, & Shaw, 2006).

BIS/BAS

Behavioral neuroscience (Gray, 1981, 1990) proposes two separate brain mechanisms that are differentially responsible for sensitivity to rewards and punishments, called the behavioral activation system (BAS) and the behavioral inhibition system (BIS). BAS regulates appetitive motivation and is activated by stimuli signaling reward (or relief from punishment), whereas BIS regulates aversive motivation and is activated by stimuli signaling punishment or frustrative nonreward (Panksepp, 2005; Weiss, 2002). Carver and White (1994) suggest that people with higher BAS sensitivity should “respond behaviorally to cues of reward and should experience positive affect in the presence of such cues,” whereas people with higher BIS sensitivity should be especially “responsive behaviorally to punishment cues and should experience great anxiety in situations with cues of impending punishment” (p. 320). BIS/BAS sensitivities as individual difference variables have been examined as antecedents of deviant work behaviors (Diefendorff & Mehta, 2007) and as moderators of within-person relationships between positive and negative mood and goal revision (Richard & Diefendorff, 2011).

Regulatory Focus

Regulatory focus theory describes how individuals regulate their behavior, proposing that self-regulation functions differently depending upon the fundamental needs underlying goal pursuit. To this end, two regulatory orientations—promotion and prevention—have been identified. Highly *promotion-focused* individuals are guided by a need for nurturance, whereas highly *prevention-focused* individuals are guided by a need for security (Higgins, 1997). Both promotion and prevention focus are approach motivations, meaning they refer to how individuals strive to achieve specific goals. Individuals with a strong promotion focus prefer to achieve their goals by maximizing positive outcomes, or “hits.” To this end, promotion focus is positively associated with the speed at which individuals work (Förster et al., 2003) and productivity in the workplace (Wallace & Chen, 2006). Conversely, individuals with a strong prevention-focus prefer to achieve their goals by minimizing mistakes, or “misses.”

Thus, a strong prevention focus is associated with accuracy (Förster et al., 2003) and safety in the workplace (Wallace & Chen, 2006; Wallace, Johnson, & Frazier, 2009).

Action-State Orientation

Action-state orientation is an individual difference variable relevant to the volitional pursuit of goals (Kuhl, 1994). Individuals with an *action orientation* are readily able to devote resources to the task at hand, whereas those with a *state orientation* “tend to have persistent, ruminative thoughts about alternative goals or affective states, which reduces the cognitive resources available for goal-striving” (Diefendorff, Hall, Lord, & Streat, 2000, p. 251). The construct of action-state orientation has three dimensions (i.e., preoccupation–disengagement, hesitation–initiative, and volatility–persistence), which relate to different aspects of the goal-striving process such that those relatively high in action orientation tend to “flexibly disengage from irrelevant concerns (preoccupation), effectively initiate required actions (hesitation), and stay focused until tasks are completed (volatility)” (Diefendorff et al., 2000, p. 251; see also Diefendorff, Richard, & Gosserand, 2006). A recent study of job search behavior by Wanberg et al. (2010) suggests action-state orientation moderates the within-person relation between lower positive affect and next-day search effort, with individuals who were more able to flexibly disengage from irrelevant concerns showing more search effort with positive affect, whereas the opposite was true for state-oriented individuals.

Goal Orientation

Another individual difference is *dispositional goal orientation*, which is the relatively stable pattern of cognition and action that results from the chronic pursuit of particular achievement goals in different situations over time (DeShon & Gillespie, 2005). The achievement goals that may be pursued include mastery/learning goals and performance goals, with each of these goals being further distinguished with regard to approach and avoidance (Baranik, Bynum, Stanley, & Lance, 2010; Hulleman et al., 2010). These are independent dimensions, such that one may tend to have a relatively high or a relatively low focus on mastery goals, on performance-approach goals and so forth (Button, Mathieu, & Zajac, 1996). A meta-analysis by Payne et al. (2007) showed relatively high test–retest reliability for dispositional goal orientation. Payne et al. also provides support for antecedents and consequences of dispositional goal orientation. Antecedents include self-esteem, implicit theories of ability (Dweck, 1986), and traits from the FFM. In particular, conscientiousness was found to relate positively to mastery/learning goals and to relate negatively to performance–avoid goals, with no significant relationship between conscientiousness and performance–approach goals. Proximal consequences of dispositional goal orientation include state goal orientation, state anxiety, and feedback seeking, whereas distal consequences include learning and performance. Dispositional goal orientation is also studied in other contexts and with other theories, including groups and teams (Porter, 2005; Porter, Webb, & Gogus, 2010), job seeking (Creed, King, Hood, & McKenzie, 2009; van Hooft & Noordzij, 2009), cross-cultural adjustment (Gong & Fan, 2006; Wang & Takeuchi, 2007), as a moderator of responses to negative feedback (Cron et al., 2005; see also Anseel et al., 2009), and as an antecedent of feedback-seeking (Park, Schmidt, Scheu, & DeShon, 2007). Further, dispositional goal orientation has been studied with regard to its interactive effects with situational goal orientation inducements on performance trajectories during skill acquisition (Chen & Mathieu, 2008).

Within-Person Variance in Individual Differences

Although individual differences are typically construed as stable over time, meaningful variance in these constructs occurs *within-individuals* over time (e.g., Fleeson, 2004, 2007; Mischel, 2004). Fleeson and Gallagher (2009) synthesized the results from 15 studies in which Big Five personality variables were measured repeatedly over time. Participants responded to Big Five adjectives (e.g., hardworking) several times per day for a period of weeks, and they also reported their standing on the Big Five *traits* by describing their behavior “in general.” Trait-level measures predicted central tendencies (e.g., mean, median, mode), indicating stable between-person differences. Yet, the majority of variance in Big Five personality variables occurred at the within-person level of analysis (extraversion, 78%; agreeableness, 63%; conscientiousness, 75%; emotional stability, 66%), with the exception of intellect (i.e., openness to experience, 49%). There is evidence for within-person variance in many other constructs related to the self and personality, including self-esteem (e.g., de Cremer, van Knippenberg, van Knippenberg, Mullenders, & Stinglhamber, 2005), goal orientation (e.g., DeShon & Gillespie, 2005; Yeo, Loft, Xiao, & Kiewitz, 2009), and many others. Within-person fluctuations in personality variables can be reliably predicted from theoretically relevant environmental conditions (Beck & Schmidt, 2011; Fleeson, 2001; Jagacinski, Kumar, Boe, Lam, & Miller, 2010; Senko & Harackiewicz, 2005; Wallace & Chen, 2006), and also predict subsequent goals and behavior (Beck & Schmidt, 2011; Yeo et al., 2009). The study of within-person variance and state-level constructs may provide an important bridge between higher level needs, goals, and the like, and more proximal and concrete goals and behaviors (e.g., Breland & Donovan, 2005; Chen et al., 2000). They may also provide points of intervention for leaders and managers seeking to influence follower and/or employee behavior (e.g., Dragoni, 2005; Kark & Van Dijk, 2007; Lord & Brown, 2004).

Temporal Dynamics

Time is an extremely important variable for work motivation (Mitchell & James, 2001). Goals must frequently be met within deadlines. Thus, the time available to complete a goal has implications for the actual and perceived difficulty of meeting the goal. In this section, we review three specific time-related topics: deadlines, procrastination, and the planning fallacy.

Deadlines

Goal assignments often incorporate deadlines, either explicitly or implicitly, for the completion of the assigned goal (Locke & Latham, 1990). Shorter deadlines often result in greater difficulty achieving goals, thereby requiring one to work at a faster pace relative to a more lax deadline (Austin & Vancouver, 1996; Locke & Latham, 1990). Given their need for immediate attention, proximal deadlines create a sense of urgency and thus increase commitment to the goal (Klein, Austin, & Cooper, 2008; Mitchell, Harman, Lee, & Lee, 2008; Waller, Conte, Gibson, & Carpenter, 2001). Steel and König's (2006) Temporal Motivation Theory (TMT) is a recent attempt to more explicitly incorporate deadlines into theories of motivation. Like traditional expectancy-value theories (e.g., Vroom, 1964), TMT proposes that the likelihood of selecting a particular course of action increases as both expectancy and the subjective value of success increase. However, TMT further proposes that the attractiveness of positive outcomes and the aversiveness of negative outcomes decrease as their occurrence moves further into the future (Ainslie, 1992). Thus, TMT predicts individuals will be drawn toward activities providing immediate or near-term benefits over

outcomes providing similar or even greater benefits that will not be realized until the future. This suggests that tasks with more proximal deadlines, as well as various “background temptations” that offer immediate benefits (e.g., socializing), will frequently command our attention, potentially to the neglect of otherwise more important tasks with later deadlines. TMT suggests that more difficult goals may be undertaken sooner than easy goals, due to greater value attached to attainment of difficult goals (Bandura, 1997). Moreover, TMT proposes that breaking a distal deadline into a series of subgoals with earlier deadlines may also reduce the likelihood of delaying the task (see also Latham & Seijts, 1999).

Whereas the above discussion highlights the potential benefits of shorter deadlines and, by extension, a greater sense of urgency or time pressure, there are also potential costs. A qualitative study by Amabile, Hadley, and Kramer (2002) suggests time pressure impairs creativity, which Lord et al. (2010) suggested may be a result of narrowed attentional focus and systematic processing evoked by anxiety. However, highlighting the potential complexity of this relationship, Amabile et al. also noted that time pressure may facilitate creativity when individuals can concentrate their efforts for a substantial portion of the day and believe the work they are doing is important and meaningful. They also note that a complete absence of time pressure may lead to insufficient engagement, with creativity suffering as a result (see also Baer & Oldham, 2006).

Procrastination

Procrastination—“to voluntarily delay an intended course of action despite expecting to be worse off for the delay” (Steel, 2007, p. 66)—is a pervasive problem (Steel, 2007, 2010). As discussed previously, Steel and König's (2006) TMT predicts that the utility of outcomes occurring in the distant future are discounted relative to the utility of outcomes occurring in the near future (i.e., hyperbolic discounting), often resulting in otherwise more important activities being put off for the future to focus on those providing more immediate benefits. However, McCrea, Liberman, Trope, and Sherman (2008) demonstrated that prompting individuals to think concretely about a behavior can lead the behavior to be started and completed with less delay than when individuals think about the same behavior in an abstract way. TMT also posits individual differences in the tendency to procrastinate. Steel's (2007) meta-analysis revealed a moderate to strong positive relationship among procrastination and impulsiveness, proneness to boredom, and distractibility. Likewise, negative relationships of procrastination with conscientiousness and need for achievement emerged. Procrastination was negatively correlated with a range of academic performance criteria. Thus, procrastination (or lack thereof) seems to be an important mediator in the relationship between non-cognitive constructs and task performance, which has been drawn upon to bolster the case for the use of noncognitive predictors in personnel selection (e.g., Ones, Dilchert, Viswesvaran, & Judge, 2007).

Similarly, Waller et al. (2001) proposed that individual differences in time urgency may influence the extent to which behavior is determined by deadlines, and combines with individual differences in future versus present time perspective to determine preferences for work pacing. Situational factors, such as proximity to deadlines and stable versus changing deadlines, can also influence time monitoring (Waller, Zellmer-Bruhn, & Giambatista, 2002). Yet, it can be difficult to predict reactions to approaching deadlines without considering the progress that one has made, and the progress that remains for goal attainment. For example, little time pressure or urgency is likely to be experienced as a deadline draws near if one has already accomplished the task at hand; yet, the same time remaining may be quite daunting when substantial work remains to be done. Consistent with this notion, Williams et al. (2000) and Donovan and Williams (2003) found that the relationship of goal-performance discrepancies on subsequent goal revision among elite collegiate athletes depended upon the time that remained in their season. Discrepancies had less impact on goal revision earlier in the season, where ample time remained to reduce the discrepancy by increasing performance, whereas the athletes tended to resolve discrepancies later in the season by bringing their goals into alignment with their performance.

Planning Fallacy

People tend to underestimate the amount of time it will take to complete tasks, a phenomenon termed the “planning fallacy.” A series of studies conducted by Buehler, Peetz, and Griffin (2010) indicate that this planning fallacy is more likely to occur for open tasks—those completed over multiple occasions and/or locations—than for closed tasks—those completed during one occasion. Furthermore, the amount of time participants predicted an open task would take to complete (a school assignment [Study 4a] or filing an income tax return [Study 4b]) predicted when participants started the task, but not when they finished. Thus, when interruptions can occur, delays in task completion can occur despite the best of intentions. Yet, Kruger and Evans (2004) demonstrated that by asking individuals to “unpack” tasks (identify the subtasks that comprise the total task), the planning fallacy was significantly reduced. In other words, across a wide variety of tasks (holiday shopping, preparing for a date, formatting a text document, and preparing a meal), individuals were more accurate in predicting completion times when tasks were unpacked. Therefore, “unpacking” may facilitate performance in jobs that require completing tasks over separate occasions under autonomous conditions.

Multiple Goals and Decision Making

We have sought to convey how work motivation reflects a process of allocating finite resources, such as time, to goals over time. Because resources like time and attention are finite, decisions must be made on a moment-to-moment basis about how to allocate these limited resources to multiple, competing goals. In this section, we attempt to tie the previous sections of this review together by focusing on theoretical and empirical work regarding resource allocation across multiple competing goals. In some ways, this section mirrors previous sections, discussing factors like goal progress, expectancy, valence, affect, and automatic goal processes. However, in this section we focus explicitly on how these factors come together to determine how individuals allocate resources among competing goals. We conclude this section with a brief review of several applications of multiple-goal self-regulation.

Theoretical and Empirical Work on Resource Allocation Processes

Goals and Goal/Performance Discrepancies

Theory and research on multiple goals is often rooted in a control theory perspective, especially in the context of work motivation (e.g., Mitchell et al., 2008). In line with this view, a series of studies by Schmidt and colleagues found that, when all else is equal (e.g., valence, expectancy, etc.), individuals often allocate more time to those goals that are most in need—that is, toward goals with larger goal-performance discrepancies (e.g., Schmidt & DeShon, 2007; Schmidt & Dolis, 2009; Schmidt, Dolis, & Tolli, 2009). Interestingly, Schmidt and DeShon (2007) also found that the tendency to favor the most discrepant goal weakens as the deadline approaches, potentially giving way to the goal *closest* to completion as time runs out. Schmidt and Dolis (2009) proposed that changes in allocation strategy may result from changes in *dual-goal expectancy*—the belief that *both* goals

can be met in the available time. When individuals believed both goals could be met, they tended to allocate more resources to whichever task was in greatest need; however, when dual-goal expectancy was low, individuals tended to favor the goal that was most likely to be met by the deadline. Schmidt et al. (2009) replicated these findings under conditions of high environmental volatility, whereby progress on both goals was influenced by unpredictable external forces in addition to the performers' own actions. However, when goal progress was determined solely by the individual's actions, effort was largely focused on one task until its completion, upon which resources were reallocated to the remaining task.

The feedback individuals receive is also critically important, as it facilitates monitoring of discrepancies. In a study of tradeoffs between individual and team performance, DeShon, Kozlowski, Schmidt, Milner, and Wiechmann (2004) found that provision of individual-level feedback resulted in greater self-focused effort and individual performance, whereas team feedback resulted in more team-focused effort and greater team performance. Provision of both individual and team feedback resulted in intermediate levels of both, highlighting the difficulty of maximizing both individual and team performance simultaneously. Similarly, Northcraft, Schmidt, and Ashford (2011) found that individuals allocated more time toward tasks providing frequent and specific feedback than tasks providing less frequent or vague feedback. Thus, feedback appears to be a valuable lever for influencing prioritization across multiple demands.

Expectancy and Valence

Consistent with expectancy theory (e.g., Vroom, 1964), an early study of multiple-goal pursuit by Kernan and Lord (1990) found that individuals allocated more time to goals with higher valences and expectancies. Likewise, Schmidt and DeShon (2007) found that, when an incentive was offered for only one of the two tasks in their study (i.e., one task had higher valence), more time was allocated toward the incentivized task. Further, consistent with control theory propositions, they found that progress on the rewarded task was more predictive of time allocation than progress on the unrewarded task. Consistent with the frequent finding that losses loom larger than gains (e.g., Kahnemann & Tversky, 1984), Schmidt and DeShon also found individuals spent more time on a task for which failure incurred a loss of a \$10 gift certificate that was provided at the beginning of the study than on the task for which success resulted in gaining an equivalent reward. Additionally, progress toward the loss-framed goal was a stronger predictor of time allocation than progress toward the gain-framed goal.

Additional studies have further demonstrated the importance of expectancy in multiple-goal self-regulation. As noted previously, expectancies regarding the likelihood of attaining *both* goals being pursued have been found to influence resource allocation strategies (Schmidt & Dolis, 2009; Schmidt et al., 2009). Louro, Pieters, and Zeelenberg (2007) showed that expectancies may mediate an interaction between goal-related emotions and goal proximity on effort allocated to competing goals. As previously discussed, Temporal Motivation Theory (Steel & König, 2006) proposes that the likelihood of engaging in a particular course of action increases with greater expectancy and value. However, they further propose that the attractiveness of positive outcomes and aversiveness of negative outcomes decrease as their occurrence moves further into the future, such that individuals are often biased toward engaging in activities with near-term consequences at the expense of those with long-term implications.

Vancouver, Weinhardt, and Schmidt (2010) developed a computational model of multiple-goal pursuit that integrates core elements of decision-making theories—drawing in particular upon TMT—with dynamic self-regulatory theories (e.g., control theory). Their model deviates from traditional expectancy-value theories primarily by explicitly specifying how expectancy and value change over the course of goal pursuit. In particular, they propose that expectancy at a given point in time—construed as the perceived likelihood of meeting a given goal by the deadline—is determined by comparing the current discrepancy to the perceived pace at which one can work. They further proposed that the subjective value of a task at a given moment is determined

not only by the consequences of success or failure, but also by the magnitude of the discrepancy, with greater “need to act” on a goal with a large discrepancy, and little or no need to act on a goal with no discrepancy. Once derived, the dynamic expectancy and value constructs are posited to combine multiplicatively, such that the task with the highest multiplicative combination of dynamic expectancy and value will be pursued at a given moment.

Complex though it is, the model contains a number of initial simplifying assumptions to be evaluated and elaborated upon in subsequent work. Nonetheless, the proposed version of the model produced simulated data that closely matched the results reported by Schmidt and DeShon (2007), including the tendency to allocate more time to the task with the largest discrepancy early on, but with this tendency weakening and potentially reversing as the deadline approaches. Yet, more work is needed to account for multiple-goal self-regulation across a wide range of scenarios.

Affect

Individuals also make use of affective information in allocating resources across multiple goals. Affect's influence on goal prioritization often occurs automatically, below conscious awareness (Barsade, Ramarajan, & Westen, 2009). Friedman and Förster (2010) reviewed research demonstrating that affective cues can influence how attention is allocated; affective cues signaling danger often cause attention to be restricted, which can facilitate quickly solving the problem at hand, whereas affective cues signaling a nonthreatening situation tend to broaden attention, encouraging exploratory behaviors. Louro et al. (2007) found that, when far from one's goals, high positive affect can facilitate persistence, whereas low positive affect can result in abandoning the goal in question in favor of a competing goal; in contrast, when close to one's goals, high positive affect can lead to premature disengagement. Louro et al. also found these effects were mediated by expectancy. They speculated that when things are perceived as going well, individuals are motivated to divert resources from one goal to another, an argument consistent with other theorists (e.g., Carver & Scheier, 1998). Putting this hypothesis to the test, Orehek, Bessarabova, Chen, and Kruglanski (2011) showed that affect was positively related to goal activation and intentions to complete the goal when no competing goal was present, but was negatively related to activation and intentions when a competing goal was present.

Nonconscious Goal Activation and Inhibition

The automatic self-regulatory mechanisms reviewed earlier have evolved precisely to help individuals manage multiple goals (Bargh, 2008). If self-regulation could occur only consciously, individuals would quickly become over-burdened by information processing requirements (Kanfer & Ackerman, 1989; Lord & Levy, 1994; R. E. Johnson et al., 2006). Thus, much of this activity happens automatically, below conscious awareness. As we have detailed earlier: (a) When pursuing important goals, competing goals are automatically inhibited from activation (R. E. Johnson et al., 2006; Shah et al., 2002); (b) the means of achieving goals are automatically activated when a higher-order goal is activated (Aarts & Dijksterhuis, 2000); and (c) information relevant to goal pursuit is more likely to be attended to in one's environment (Vogt et al., 2010) and is more likely to be accessed in working memory (R. E. Johnson et al., 2006). Thus, automatic self-regulatory processes help individuals efficiently manage multiple competing goals without overburdening finite resources.

Applied Examples of Multiple-Goal Self-Regulation in the Workplace

Balancing multiple goals is a common activity in the workplace. We review some of the most common multiple-goal situations that appear in the work motivation literature and consider how the research reviewed above can inform each topic.

Speed Versus Accuracy

A prevalent trade-off in the workplace is between performing a task quickly and performing a task accurately (i.e., quantity vs. quality). Gilliland and Landis (1992) showed that such trade-offs were most likely when the task being performed was difficult, as participants could not readily meet both goals. Locke, Smith, Erez, Chah, and Schaffer (1994) found participants could be instructed to emphasize either quantity or quality, which resulted in corresponding patterns of commitment and performance across quality and quantity aspects of the task. Förster et al. (2003) found those with a strong promotion focus tend to favor speed over accuracy, yet those with a strong prevention focus tend to favor accuracy. Managing speed and accuracy seems to be a core process, with deep evolutionary roots (Chittka, Skorupski, & Raine, 2009), that is critical to successful navigation of a wide variety of environments.

Safety Versus Efficiency

Many tasks can be completed more quickly if safety procedures are not followed (e.g., Wallace & Chen, 2006; Weyman & Clarke, 2003); however, the consequences of this approach can be severe (Blount, Waller, & Leroy, 2005). To date, research integrating motivation and safety has emphasized a between-person and between-group perspective (e.g., Christian, Bradley, Wallace, & Burke, 2009; Nahrgang, Morgeson, & Hofmann, 2011), in which individual differences and environmental characteristics are correlated with aggregated safety outcomes (e.g., supervisor ratings, injuries, accidents) over a period of time. However, research is needed to understand how individuals make trade-offs between safety and efficiency on a moment-by-moment basis, which may provide insights into interventions that may be useful for influencing safety behaviors at a given moment in time.

Development Versus Short-Term Performance

Because development takes time, there is often a trade-off between short-term performance and longer-term developmental goals. In a study of salespeople immediately following the introduction of new software, Ahearne, Lam, Mathieu, and Bolander (2010) found that individuals with high mastery goal orientations (MGOs) initially experienced decreased sales performance, presumably due to spending time learning the new software rather than focusing on their sales. Yet, these individuals eventually improved their performance over baseline, whereas those with low MGOs never returned to pre-intervention performance levels. The opposite pattern emerged for performance-prove (PGO) goal orientation, as those concerned with proving their abilities to others experienced less initial drop in sales performance, but never returned to their baseline levels. Individuals may be more willing to focus on development when they are performing well (Senko & Harackiewicz, 2005), believe they can develop their skills (Jagacinski et al., 2010), have low fear of failure (Elliot & Fryer, 2008), and perceive low time pressure (Beck & Schmidt, 2011). Dragoni (2005) suggests managers can create climates for development, performance, and avoiding failure via their own

patterns of achievement motivation. Gregory, Beck, and Carr (2011; see also Beck, Gregory, & Carr, 2009) suggest coaching relationships may provide opportunities to help employees strike the appropriate balance between short- and long-term performance via developmental pursuits.

Ethical Decision Making

Individuals may be able to maximize some goals (e.g., maximizing rewards and recognition) by sacrificing ethical goals. In a recent exchange in the *Academy of Management Perspectives*, Ordóñez, Schweitzer, Galinsky, and Bazerman (2009) postulated that assignment of difficult-specific goals may promote unethical behaviors undertaken to achieve them. They noted that difficult goals tend to (a) narrow one's focus to the task at hand (to the neglect of other concerns: e.g., Shah et al., 2002), (b) lead to a focus on short-term gains instead of long-term implications (e.g., Camerer, Babcock, Loewenstein, & Thaler, 1997), and (c) increase acceptance of risky behavior (e.g., Larrick, Heath, & Wu, 2009). Schweitzer, Ordóñez, and Douma (2004) found participants were more likely to overstate their performance when given difficult-specific goals, particularly when their performance fell just short of the goal. Locke and Latham (2009) countered that much of Ordóñez et al.'s arguments were based on anecdotal evidence, and dismissed the Schweitzer et al. study as an aberration, citing a large body of literature demonstrating the positive effects of goal setting. Nonetheless, we believe more research on this issue will emerge in the coming years, furthering understanding of when and why “goals go wild.” Reynolds (2006) suggests that, whereas many ethical decisions are well practiced and performed automatically without conscious awareness, others are novel and would benefit from more controlled processing. Similarly, Barnes, Schaubroeck, Huth, and Ghumman (in press) showed that lack of sleep led to a depletion of regulatory resources (i.e., cognitive fatigue), which in turn resulted in more unethical behavior. More research is needed to elaborate on these processes as well as to provide potential remedies.

Work–Life Conflict

Individuals often must balance work goals with nonwork goals. Much of this research focuses on antecedents (e.g., hours spent at work, job stressors, supportive work environment, familial support) and outcomes (e.g., job satisfaction, family satisfaction, stress, health) of work–life conflict (e.g., Ford, Heinen, & Langkamer, 2007; Mesmer-Magnus & Viswesvaran, 2005), but give little attention to how individuals seek to balance these concerns when conflict is perceived. We believe that a multiple-goal perspective may prove beneficial. For one, goals relating to one's personal life (e.g., child care, hobbies) may often reside higher in the goal hierarchy than work goals and thus may provide more motivational “pull” than work goals. Further, in some cases personal life goals may be construed as things the individual “ought to do,” meaning work goals are likely to be shielded from these personal goals. Insights from multiple-goal self-regulation may bolster existing organizational interventions seeking to minimize work–life conflict, and may also lead to additional approaches yet to be considered.

Discussion

Summary

As our review indicates, work motivation is a constellation of dynamic, reciprocal processes that unfold over time. At the core of such processes are goals, the desired states human beings strive to achieve. These goals vary considerably in their content, level

of specificity, importance, and time frames over which they are pursued. Some goal processes occur consciously, such as choosing one course of action over another. However, many of these processes happen below conscious awareness, guiding behavior without the burden of conscious thought and attention. Goal pursuit can further be defined as the allocation of resources. That is, to pursue a goal an individual must allocate his or her resources, be it time, effort, money, and so on. Likewise, given the finite nature of most resources, the decision to allocate to one goal is often implicitly the decision not to allocate to another. Goal pursuit is supported by a variety of other psychological processes, such as projections about one's chances of success (self-efficacy, expectancy, etc.), subjective feelings of "rightness" and "wrongness" (e.g., value from regulatory fit), and pleasure derived from successful goal pursuit (e.g., positive affect). Although goal pursuit is a human universal, individual differences such as personality and disposition reflect preferences in *what* goals to pursue or *how* to pursue them. Thus, the study of work motivation is the study of the allocation of resources to goals over time—those pertaining to the work itself as well as the myriad goals an individual pursues in tandem with work-related goals—along with the processes that accompany goal pursuit.

Future Directions

Despite the substantial progress that has been made in recent years, much remains to be learned regarding motivation and self-regulation in the workplace. Here, we briefly highlight a few, among many, issues we believe warrant additional consideration in future research. First, although organizational scholars have recently begun taking the automatic/unconscious more seriously, we believe this movement has only just begun. A large literature exists within cognitive and social-cognitive psychology regarding these issues. Although important strides have been made in extending this research to the organizational domain, we believe further application will likely prove beneficial. Further, in addition to co-opting existing concepts from other areas of psychology, organizational scholars have many opportunities to contribute to the broader field of psychology with regard to this boundless domain of inquiry. For instance, an issue of great practical relevance in the workplace concerns the "durability" of priming effects—that is, how long do subtle primes continue to exert an effect, particularly in the face of the constant barrage of potential counter-primes individuals are likely to face in the work environment. Similarly, more needs to be known concerning how competing primes are reconciled. Additionally, it would likely prove beneficial to better understand the trade-offs between conscious/effortful and automatic/mindless processes in the workplace. That is, under what conditions should automatic processes be favored over more mindful processing, and vice versa? This is, in large part, an issue of how to best utilize individuals' limited capacity for effortful, conscious processing. Finally, given the historical interest and expertise in individual differences, I-O psychologists may be uniquely poised to contribute to knowledge concerning the role of individual differences in priming, automaticity, and other implicit processes.

Second, as noted earlier, research on affect in the workplace has been flourishing. However, much more remains to be learned regarding how affect influences and is influenced by motivational processes. Whereas research has focused largely on positive and negative affect—and to a lesser extent the affect circumplex obtained by crossing positive and negative valence with high and low activation (Weiss, 2002)—other emotions may hold relevance for organizational behavior. For example, researchers in affective neuroscience have identified emotional systems that give rise to emotions of rage, lust, fear, care, panic, seeking, and play (e.g., Panksepp, 1998, 2010). Further, in addition to the emotional affects, there are also sensory affects and bodily–homeostatic affects. Sensory affects reflect sensory experiences ranging from pleasures to displeasure (e.g., disgust) as well as bodily disturbances (e.g., pain, fatigue), whereas bodily–homeostatic affects gauge bodily need states (e.g., hunger, and thirst; Panksepp, 2005, 2008). Relatively little work has examined the intersection of such experiences with motivational processes. Future research may also benefit from further examination of emotional labor and affect regulation in groups (George, 2002;

Pugh, 2002), including phenomena such as emotional contagion and mimicry (e.g., Barger & Grandey, 2006). There is also likely to be value in examining automatic emotion regulation processes (e.g., Lazarus, 1975; Moon & Lord, 2006). Finally, like most organizational phenomena, affect is often examined from a conscious perspective. Yet, individuals may often be unaware of the causes of their affective experiences, be unaware of the way their affective experiences influence subsequent cognition and action, and may even sometimes be unaware of the emotional experience itself (e.g., Barsade et al., 2009). This is an intriguing line of inquiry, with many implications yet to be uncovered.

Third, the study of time as a substantive issue is another area we believe continues to hold substantial promise for the future of work motivation research. Throughout this chapter, we have highlighted some of the ways time has been examined in the literature. Given its position as a key resource in organizational behavior, time is likely to remain a fruitful area of study in the future, as much remains to be learned regarding the role of time in motivational processes. For example, what is the role of perceptions of time available versus time required to meet a goal in the effects of anticipatory constructs such as self-efficacy and expectancy? What impact does affect have on perceptions of time, and how do perceptions of time and deadlines influence affect? How does the progression of time influence the reliance on and effectiveness of implicit versus explicit processes? What, if anything, can be done to reduce problems and biases associated with time and deadlines? Although existing research provides some valuable insights on these issues, further advancements are likely to provide additional practical implications.

Fourth, and perhaps most broadly, we strongly encourage additional efforts toward integrating theories of motivation. Although important strides have been made in this regard, much work remains. At present, numerous theories of motivation have held up, at least in part, to empirical scrutiny. While there can be great utility in multiple theoretical perspectives, including relatively independent “micro-theories” regarding particular motivational phenomena or regarding motivation within a particular context, such an approach also presents potential for duplication or neglect of relevant existing work, confusion of terminology, conflicting propositions or results (albeit the reconciliation of which can often advance our understanding), and other such pitfalls. This call for integration is by no means new (e.g., Diefendorff & Lord, 2008; Donovan, 2001; Kanfer, 1990; Vancouver, 2008). However, we believe it is all the more important as the motivational sciences continue to mature and expand their focus.

Given the complexity of the phenomena involved, such integrative efforts are likely to benefit greatly from increased utilization of computational modeling (Ilgen & Hulin, 2000; Vancouver, 2008; Vancouver, Putka, & Scherbaum, 2005). Computational modeling can be a vital tool for integrating theories of motivation for several reasons. Computational modeling necessitates that relationships be expressed in concrete, mathematical forms. This explicitness helps researchers to communicate among each other regarding the exact nature of the phenomenon under investigation, as well as to evaluate much more concretely whether a set of empirical observations matches what was hypothesized. By replacing subjective language with objective mathematical formulas, computational modeling will help researchers avoid problems such as classifying the same motivational phenomena under different names or utilizing the same label for distinct constructs, relying upon unstated and potentially unrecognized assumptions, misinterpreting authors' intended meaning due to differences in language or even from differences in theoretical or disciplinary background, and so on. Computational modeling also allows one to evaluate whether a particular theoretical account is indeed capable of reproducing the phenomenon in question. Although a theory is ultimately evaluated against genuine observations, failure of the model to replicate the phenomenon of interest suggests the need to revise one's theoretical account, which may be a highly valuable driver of theoretical development. An additional benefit of computational modeling is that complex, multivariate, reciprocal processes can be examined without the human information processing limitations involved in mental simulation and other such nonmathematical approaches, which can be subject to a variety of errors and biases. Particularly with complex and highly dynamic theories, computational models may provide insights and generate predictions unlikely to have been obtained otherwise.

Conclusion

As we hope this review shows, work motivation remains a very active field of research within industrial–organizational psychology and organizational behavior. Although much remains to be learned, the research to date provides a wealth of information regarding motivational antecedents, processes, and outcomes. This knowledge has been fruitfully applied to beneficial effect, and we believe the practical benefits of this body of research will continue to grow as our knowledge expands.

References

- Aarts, H., & Dijksterhuis, A. (2000). Habits as knowledge structures: Automaticity in goal-directed behavior. *Journal of Personality and Social Psychology*, 78, 53–63.
- Ahearne, M. Lam, S. K., Mathieu, J. E., & Bolander, W. (2010). Why are some salespeople better at adapting to organizational change? *Journal of Marketing*, 74, 65–79.
- Ahearne, M., Mathieu, J., & Rapp, T. (2005). To empower or not to empower your sales force? An empirical examination of the influence of empowering leader behaviors on customer satisfaction and performance. *Journal of Applied Psychology*, 90, 5, 735–744.
- Ainslie, G. (1992). *Picoeconomics: The strategic interaction of successive motivational states within the person*. New York, NY: Cambridge University Press.
- Allport, G. W. (1955). *Becoming*. New Haven, CT: Yale University Press.
- Allport, G. W. (1961). *Pattern and growth in personality*. New York, NY: Holt, Rinehart & Winston.
- Amabile, T., Hadley, C. N., & Kramer, S. J. (2002). Creativity under the gun. *Harvard Business Review*, 80, 52–61.
- Ames, D. R. (2008). Assertiveness expectancies: How hard people push depends on the consequences they predict. *Journal of Personality and Social Psychology*, 95, 1541–1557.
- Anseel, F., Lievens, F., & Schollaert, E. (2009). Reflection as a strategy to enhance task performance after feedback. *Organizational Behavior and Human Decision Processes*, 110, 23–35.
- Austin, J. T., & Vancouver, J. B. (1996). Goal constructs in psychology: Structure, process and content. *Psychological Bulletin*, 120, 338–375.
- Baer, M., & Oldham, G. R. (2006). The curvilinear relation between experienced creative time pressure and creativity: Moderating effects of openness to experience and support for creativity. *Journal of Applied Psychology*, 91, 963–970.
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Englewood Cliffs, NJ: Prentice-Hall.
- Bandura, A. (1997). *Self-efficacy: The exercise of control*. New York, NY: Freeman.
- Bandura, A. (2006). Toward a psychology of human agency. *Perspectives on Psychological Science*, 1, 164–180.
- Bandura, A., & Locke, E. (2003). Negative self-efficacy and goal effects revisited. *Journal of Applied Psychology*, 88, 87–99.
- Baranik, L. E., Bynum, B. H., Stanley, L. J., & Lance, C. E. (2010). Examining the construct validity of mastery–avoidance achievement goals: A meta-analysis. *Human Performance*, 23, 265–282.

- Barger, P., & Grandey, A. (2006). Service with a smile and encounter satisfaction: Emotional contagion and appraisal mechanisms. *Academy of Management Journal*, 49, 1229–1238.
- Bargh, J. A. (2008). Free will is un-natural. In J. Baer, J. C. Kaufman, & R. F. Baumeister (Eds.), *Are we free? Psychology and free will* (pp. 128–154). New York, NY: Oxford University Press.
- Bargh, J. A., & Chartrand, T. L. (2000). The mind in the middle: A practical guide to priming and automaticity research. In H. T. Reis & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 253–285). New York, NY: Cambridge University Press.
- Barnes, C. E., Schaubroeck, J., Huth, M., & Ghumman, S. (in press). Lack of sleep and unethical behavior. *Organizational Behavior and Human Decision Processes*.
- Barrick, M. R., & Mount, M. K. (1991). The Big Five personality dimensions and job performance: A meta-analysis. *Personnel Psychology*, 44, 1–26.
- Barrick, M. R., Mount, M. K., & Judge, T. A. (2001). Personality and performance at the beginning of the new millennium: What do we know and where do we go next? *International Journal of Selection & Assessment*, 9, 9–30.
- Barrick, M. R., Parks, L., & Mount, M. K. (2005). Self-monitoring as a moderator of the relationships between personality traits and performance. *Personnel Psychology*, 58, 745–768.
- Barsade, S., Ramarajan, L., & Westen, D. (2009). Implicit affect in organizations. In B. M. Staw and A. P. Brief (Eds.), *Research in organizational behavior*, 29, 135–162.
- Barsade, S. G., & Gibson, D. E. (2007). Why does affect matter in organizations? *Academy of Management Perspectives*, 21, 36–59.
- Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, 117, 497–529.
- Baumeister, R. F. (1998). The self. In D. T. Gilbert, S. T. Fiske, & G. Lindzey (Eds.), *The handbook of social psychology* (4th ed., pp. 680–740). Boston, MA: McGraw-Hill.
- Beal, D. J., Weiss, H. M., Barros, E., & MacDermid, S. M. (2005). An episodic process model of affective influences on performance. *Journal of Applied Psychology*, 90, 1054–1068.
- Beck, J. W., & Schmidt, A. M. (2011a, April). A longitudinal study of goal orientations, time pressure, and performance. Poster presented at the 26th Annual Conference of the Society for Industrial and Organizational Psychologists, Chicago, IL.
- Beck, J. W., & Schmidt, A. M. (2011b, August). Dynamic test-taking motivation: Effects of self-efficacy on time allocation and performance. Paper presented at the 71st Annual Conference of the Academy of Management, San Antonio, TX.
- Beck, J. W., & Schmidt, A. M. (2011c, April). Location-location-location! Between-person self-efficacy moderates within-person effects on effort and performance. Poster presented at the 26th Annual Conference of the Society for Industrial and Organizational Psychologists, Chicago, IL.
- Beck, J. W., Gregory, J. B., & Carr, A. E. (2009). Balancing development with day-to-day task demands. A multiple-goal approach to executive coaching. *Industrial and Organizational Psychology: Perspectives on Science and Practice*, 2, 293–296.
- Blackwell, L. S., Trzesniewski, K. H., & Dweck, C. S. (2007). Implicit theories of intelligence predict achievement scores across an adolescent transition: A longitudinal study and an intervention. *Child Development*, 78, 246–263.

- Blount, S., Waller, M., & Leroy, S. (2005). Coping with temporal uncertainty: When rigid, ambitious deadlines don't make sense. In W. Starbuck, & M. Farjoun (Eds.), *Organization at the limit* (pp. 122–139). Oxford, England: Blackwell.
- Bono, J. E., & Vey, M. A. (2007). Personality and emotional performance: Extraversion, neuroticism, and self-monitoring. *Journal of Occupational Health Psychology*, 12, 177–192.
- Breland, B. T., & Donovan, J. J. (2005). The role of state goal orientation in goal establishment process. *Human Performance*, 18, 23–53.
- Brief, A. P., & Weiss, H. M. (2002). Organizational behavior: Affect in the workplace. *Annual Review of Psychology*, 53, 279–307.
- Brotheridge, C. M., & Grandey, A. A. (2002). Emotional labor and burnout: Comparing perspectives of “people work.” *Journal of Vocational Behavior*, 60, 17–39.
- Brunstein, J., Schultheiss, O., & Grassmann, R. (1998). Personal goals and emotional well-being: The moderating role of motive dispositions. *Journal of Personality and Social Psychology*, 75, 494–508.
- Buehler, R., Peetz, J., & Griffin, D. (2010). Finishing on time: When do predictions influence completion times? *Organizational Behavior and Human Decision Processes*, 111, 23–32.
- Button, S. B., Mathieu, J. E., & Zajac, D. M. (1996). Goal orientation in organizational research: A conceptual and empirical foundation. *Organizational Behavior & Human Decision Processes*, 67, 26–48.
- Camerer, C., Babcock, L., Loewenstein, G., & Thaler, R. (1997). Labor supply of New York City cabdrivers: One day at a time. *The Quarterly Journal of Economics*, 112, 407–441.
- Campbell, J. P., McCloy, R. A., Oppler, S. H., & Sager, C. E. (1993). A theory of performance. In N. Schmitt & W. C. Borman (Eds.), *Personnel selection in organizations* (pp. 35–70). San Francisco, CA: Jossey-Bass.
- Carver, C. S., & Scheier, M. F. (1998). *On the self-regulation of behavior*. New York, NY: Cambridge University Press.
- Carver, C. S., & Scheier, M. F. (2008). *Perspectives on personality* (6th ed.) Needham Heights, MA: Allyn & Bacon.
- Carver, C. S., & White, T. L. (1994). Behavioral inhibition, behavioral activation, and affective responses to impending reward and punishment: The BIS/BAS scales. *Journal of Personality and Social Psychology*, 67, 319–333.
- Chang, C-H., Johnson, R. E., & Lord, R. G. (2010). Moving beyond discrepancies: The importance of velocity as a predictor of job satisfaction and motivation. *Human Performance*, 23, 58–80.
- Chau, S. L., Dahling, J. J., Levy, P. E., & Diefendorff, J. M. (2009). A predictive study of emotional labor and turnover. *Journal of Organizational Behavior*, 30, 1151–1163.
- Chen, G., Gully, S. M., Eden, D. (2001). Validation of a new general self-efficacy scale. *Organizational Research Methods*, 4, 62–83.
- Chen, G., Gully, S. M., Whiteman, J. A., & Kilcullen, B. N. (2000). Examination of relationships among trait-like individual differences, state-like individual differences, and learning performance. *Journal of Applied Psychology*, 85, 834–847.
- Chen, G., Kanfer, R., DeShon, R. P., Mathieu, J. E., & Kozlowski, S. W. J. (2009). The motivating potential of teams: A test and extension of Chen & Kanfer's (2006) model. *Organizational Behavior and Human Decision Processes*, 110, 45–55.
- Chen, G., & Mathieu, J. E. (2008). Goal orientation dispositions and performance trajectories: The roles of supplementary and complementary situational inducements. *Organizational Behavior and Human Decision Processes*, 106, 21–38.

- Chen, G., Ployhart, R. E., Thomas, H. C., Anderson, N., & Bliese, P. D. (2011). The power of momentum: A new model of dynamic relationships between job satisfaction change and turnover intentions. *Academy of Management Journal*, 54, 159–181.
- Chen, G., Thomas, B., & Wallace, C. J. (2005). A multilevel examination of the relationships among training outcomes, mediating regulatory processes, and adaptive performance. *Journal of Applied Psychology*, 90, 827–841.
- Chittka, L., Skorupski, P., & Raine, N. E. (2009). Speed-accuracy tradeoffs in animal decision making. *Trends in Ecology and Evolution*, 24, 400–407.
- Christian, M. S., Bradley, J. C., Wallace, J. C., & Burke, M. J. (2009). Workplace safety: A meta-analysis of the roles of person and situation factors. *Journal of Applied Psychology*, 94, 1103–1127.
- Cianci, A. M., Klein, H. J., & Seijts, G. H. (2010). The effect of negative feedback on tension and subsequent performance: The main and interactive effects of goal content and conscientiousness. *Journal of Applied Psychology*, 95, 618–630.
- Colbert, A. E., & Witt, L. A. (2009). The role of goal-focused leadership in enabling the expression of conscientiousness. *Journal of Applied Psychology*, 94, 790–796.
- Colquitt, J. A., Scott, B. A., Judge, T. A., & Shaw, J. C. (2006). Justice and personality: Using integrative theories to derive moderators of justice effects. *Organizational Behavior and Human Decision Processes*, 100, 110–127.
- Converse, P. D., Steinhauser, E., & Pathak, J. (2010). Individual differences in reactions to goal-performance discrepancies over time. *Personality and Individual Differences*, 48, 138–143.
- Conway, J., Rogelberg, S. G., Pitts, V. (2009). Workplace helping: Interactive effects of personality and momentary positive affect. *Human Performance*, 22, 321–339.
- Côté, S., & Morgan, L. M. (2002). A longitudinal analysis of the association between emotion regulation, job satisfaction, and intentions to quit. *Journal of Organizational Behavior*, 23, 947–962.
- Creed, P. A., King, V., Hood, M., & McKenzie, R. (2009). Goal orientation, self-regulation strategies, and job-seeking intensity in unemployed adults. *Journal of Applied Psychology*, 94, 806–813.
- Cron, W., Slocum, J., VandeWalle, D., & Fu, Q. (2005). The role of goal on negative emotions and goal setting when initial performance falls short of one's performance goal. *Human Performance*, 18, 55–80.
- Cury, F. Elliot, A., Sarrazin, P., Da Fronseca, D., & Rufo, M. (2002). The trichotomous achievement goal model and intrinsic motivation: A sequential meditational analysis. *Journal of Experimental Social Psychology*, 38, 473–481.
- Dalal, R. S., & Hulin, C. L. (2008). Motivation for what? The criterion question. In R. Kanfer, G. Chen, & R. D. Pritchard (Eds.), *Work motivation: Past, present and future* (pp. 63–100). New York, NY: Routledge.
- Dalal, R. S., Lam, H., Weiss, H. M., Welch, E., & Hulin, C. L. (2009). A within-person approach to work behavior and performance: Concurrent and lagged citizenship-counterproductivity associations, and dynamic relationships with affect and overall job performance. *Academy of Management Journal*, 52, 1051–1066.
- de Cremer, D., Van Knippenberg, B., Van Knippenberg, D., Mullenders, D., & Stinglhamber, F. (2005). Rewarding leadership and fair procedures as determinants of self-esteem. *Journal of Applied Psychology*, 90, 3–12.
- Deci, E. L., & Ryan, R. M. (2000). The “what” and “why” of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, 11, 227–268.
- DeShon, R. P., & Gillespie, J. Z. (2005). A motivated action theory account of goal orientation. *Journal of Applied Psychology*, 90, 1096–1127.

- DeShon, R. P., Kozlowski, S. W. J., Schmidt, A. M., Milner, K. R., & Wiechmann, D. (2004). Multiple goal feedback effects on the regulation of individual and team performance in training. *Journal of Applied Psychology*, 89, 1035–1056.
- Dickhäuser, O., & Reinhard, M. (2006). Factors underlying expectancies of success and achievement: the influential roles of need for cognition and general or specific self-concepts. *Journal of Personality and Social Psychology*, 90, 490–500.
- Diefendorff, J. M., & Chandler, M. M. (2010). Motivating employees. In S. Zedeck (Ed.). *Handbook of industrial and organizational psychology* (pp. 65–135). Washington, DC: American Psychological Association.
- Diefendorff, J. M., & Gosserand, R. H. (2003). Understanding the emotional labor process: A control theory perspective. *Journal of Organizational Behavior*, 24, 945–959.
- Diefendorff, J. M., Hall, R. J., Lord, R. G., & Streat, M. L. (2000). Action-state orientation: Construct validity of a revised measure and its relationship to work-related variables. *Journal of Applied Psychology*, 85, 250–263.
- Diefendorff, J. M., & Lord, R. G. (2008). Goal-striving and self-regulation processes. In R. Kanfer, G. Chen, & R. D. Pritchard (Eds.), *Work motivation: Past, present, and future* (pp. 151–196). New York, NY: Routledge.
- Diefendorff, J. M., & Mehta, K. (2007). The relations of motivational traits with workplace deviance. *Journal of Applied Psychology*, 92, 967–977.
- Diefendorff, J. M., Richard, E. M., & Gosserand, R. (2006). Examination of situational and attitudinal moderators of the hesitation and performance relation. *Personnel Psychology*, 59, 365–393.
- Donovan, J. J. (2001). Work motivation. In N. Anderson, D. S. Ones, H. K. Sinangil, & C. Viswesvaran (Eds.), *Handbook of industrial, work, & organizational psychology* (pp. 53–76). N. London, England: Sage.
- Donovan, J. J. (2009). Antecedents of discrepancy production. *Journal of Managerial Issues*, 21, 402–420.
- Donovan, J. J., & Hafsteinsson, L. G. (2006). The impact of goal-performance discrepancies, self-efficacy and goal orientation on upward goal revision. *Journal of Applied Social Psychology*, 36, 1046–1099.
- Donovan, J. J., & Williams, K. J. (2003). Missing the mark: Effects of time and causal attributions on goal revision in response to goal-performance discrepancies. *Journal of Applied Psychology*, 88, 379–399.
- Dragoni, L. (2005). Understanding the emergence of state goal orientation in organizational work groups: The role of leadership and multilevel climate perceptions. *Journal of Applied Psychology*, 90, 1084–1095.
- Dudley, N. M., Orvis, K. A., Lebiecki, J. E., & Cortina, J. M. (2006). A meta-analytic investigation of conscientiousness in the prediction of job performance: Examining the intercorrelations and the incremental validity of narrow traits. *Journal of Applied Psychology*, 91, 40–57.
- Dweck, C. S. (1986). Motivational processes affecting learning. *American Psychologist*, 41, 1040–1048.
- Dweck, C.S. (2008). Can personality be changed? The role of beliefs in personality and change. *Current Directions in Psychological Science*, 17, 391–394.
- Eden, D. (2001). Means efficacy: External sources of general and specific subjective efficacy. In M. Erez, U. Kleinbeck, & H. Thierry (Eds.), *Work motivation in the context of a globalizing economy* (pp. 73–85). Mahwah, NJ: Erlbaum.
- Eden, D., Ganzach, Y., & Flumin-Granat, R., & Zigman, T. (2010). Augmenting means efficacy to improve performance: Two field experiments. *Journal of Management*, 36, 687–713.

- Elicker, J. D., Lord, R. G., Ash, S. R., Kohari, N. E., Hruska, B. J., McConnell, N. L., & Medvedeff, M. E. (2010). Velocity as a predictor of performance satisfaction, mental focus, and goal revision. *Applied Psychology: An International Review*, 59, 495–514.
- Elliot, A. J., & Fryer, J. W. (2008). The goal construct. In J. Shah & W. Gardner (Eds.), *Handbook of motivation science* (pp. 235–250). New York, NY: Guilford Press.
- Erez, A., & Isen, A. M. (2002). The influence of positive-affect on expectancy motivation: Integrating affect and cognition into motivation theories. *Journal of Applied Psychology*, 87, 1055–1067.
- Ferguson, M. J. (2008). On becoming ready to pursue a goal you don't know you have: Effects of nonconscious goals on evaluative readiness. *Journal of Personality and Social Psychology*, 95, 1268–1294.
- Ferris, D. L., Lian, H., Brown, D. J., Pang, F. X., & Keeping, L. M. (2011). Self-esteem and job performance: The moderating role of self-esteem contingencies. *Personnel Psychology*, 63, 561–593.
- Fitzsimons, G. M., & Finkel, E. J. (2011). Outsourcing self-regulatory effort to significant others. *Psychological Science*, 22, 369–375.
- Fleeson, W., & Gallagher, P. (2009). The implications of Big Five standing for the distribution of trait manifestation in behavior: Fifteen experience-sampling studies and a meta-analysis. *Journal of Personality and Social Psychology*, 97, 1097–1114.
- Fleeson, W. (2001). Towards a structure- and process-integrated view of personality: Traits as density distributions of states. *Journal of Personality and Social Psychology*, 80, 1011–1027.
- Fleeson, W. (2004). Moving personality beyond the person–situation debate: The challenge and the opportunity of within-person variability. *Current Directions in Psychological Science*, 13, 83–87.
- Fleeson, W. (2007). Situation-based contingencies underlying trait-content manifestation in behavior. *Journal of Personality*, 75, 825–861.
- Ford, M., Heinen, B., & Langkamer, K. (2007). Work and family satisfaction and conflict: A meta-analysis of cross-domain relations. *Journal of Applied Psychology*, 92, 57–80.
- Förster, J., Higgins, E. T., & Bianco, A. T. (2003). Speed/accuracy decisions in task performance: Built-in trade-off or separate strategic concerns? *Organizational Behavior and Human Decision Processes*, 90, 148–164.
- Förster, J., Liberman, N., & Higgins, E. T. (2005). Accessibility from active and fulfilled goals. *Journal of Experimental Social Psychology*, 41, 220–239.
- Friedman, R. S., & Förster, J. (2010). Implicit cues and attentional tuning: An integrative review. *Psychological Bulletin*, 136, 875–893.
- Gagné, M., & Deci, E. L. (2005). Self-determination theory and work motivation. *Journal of Organizational Behavior*, 26, 331–362.
- Gagné, M., & Forest, J. (2008). The study of compensation systems through the lens of self-determination theory: Reconciling 35 years of debate. *Canadian Psychology*, 49, 225–232.
- George, J. M. (2002). Affect regulation in groups and teams. In R. Lord, R. Klimoski, & R. Kanfer (Eds.), *Emotions and work* (pp. 183–217). San Francisco, CA: Organizational Frontiers Series, Jossey-Bass.
- Gillespie, J. Z., Barger, P. B., Yugo, J. E., Conley, C. J., & Ritter, L. (in press). The suppression of negative emotions in elder care. *Journal of Managerial Psychology*.

- Gilliland, S. W., & Landis, R. S. (1992). Quality and quantity goals in a complex decision task: Strategies and outcomes. *Journal of Applied Psychology*, 77, 672–681.
- Goldberg, L., & Grandey, A. A. (2007). Display rules versus display autonomy: Emotion regulation, emotional exhaustion, and task performance in a call center simulation. *Journal of Occupational Health Psychology*, 12, 301–318.
- Goncalo, J. A., Polman, E., & Maslach, C. (2010). Can confidence come too soon? Collective efficacy, conflict and group performance over time. *Organizational Behavior and Human Decision Processes*, 113, 13–24.
- Gong, Y., & Fan, J. (2006). Longitudinal examination of the role of goal orientation in cross-cultural adjustment. *Journal of Applied Psychology*, 91, 176–184.
- Gong, Y., Huang, J.-C., & Farh, J.-L. (2009). Employee learning orientation, transformational leadership, and employee creativity: The mediating role of employee creative self-efficacy. *Academy of Management Journal*, 52, 765–778.
- Gosserand, R. H., & Diefendorff, J. M. (2005). Emotional display rules and emotional labor: The moderating role of commitment. *Journal of Applied Psychology*, 90, 1256–1264.
- Grandey, A. A. (2000). Emotion regulation in the workplace: A new way to conceptualize emotional labor. *Journal of Occupational Health Psychology*, 5, 95–110.
- Grandey, A. A. (2003). When “the show must go on”: Surface acting and deep acting as determinants of emotional exhaustion as determinants of emotional exhaustion and peer-rated service delivery. *Academy of Management Journal*, 46, 86–96.
- Grandey, A. A., & Brauburger, A. L. (2002). The emotion regulation behind the customer service smile. In R. G. Lord, R. J. Klimoski, & R. Kanfer (Eds.), *Emotions in the workplace: Understanding the structure and role of emotions in organizational behavior* (pp. 260–294). San Francisco: Jossey-Bass.
- Grandey, A. A., Fisk, G. M., & Steiner, D. D. (2005). Must “service with a smile” be stressful? The moderate role of personal control for American and French employees. *Journal of Applied Psychology*, 90, 893–904.
- Gray, J. A. (1981). A critique of Eysenck’s theory of personality. In H. J. Eysenck (Ed.), *A model for personality* (pp. 246–276). New York, NY: Springer.
- Gray, J. A. (1990). Brain systems that mediate both emotion and cognition. *Cognition and Emotion*, 4, 269–288.
- Gregory, J. B., Beck, J. W., & Carr, A. E. (2011). Goals, feedback, and self-regulation: Control theory as a natural framework for executive coaching. *Consulting Psychology Journal: Practice and Research*, 63, 26–38.
- Greguras, G. J., & Diefendorff, J. M. (2009). Different fits satisfy different needs: Linking person–environment fit to employee commitment and performance using self-determination theory. *Journal of Applied Psychology*, 94, 465–477.
- Griffeth, R. W., Hom, P. W., & Gaertner, S. (2000). A meta-analysis of antecedents and correlates of employee turnover: Update, moderator tests, and research implications for the next millennium. *Journal of Management*, 26, 463–488.
- Gross, J. J. (1998a). Antecedent- and response-focused emotion regulation: Divergent consequences for experience, expression, and physiology. *Journal of Personality and Social Psychology*, 74, 224–237.
- Gross, J. J. (1998b). The emerging field of emotion regulation: An integrative review. *Review of General Psychology*, 2, 271–299.
- Grouzet, F. M. E., Kasser, T., Ahuvia, A., Dols, J. M. F., Kim, Y., Lau, S.,...Sheldon, K. M. (2005). The structure of goal contents across 15 cultures. *Journal of Personality and Social Psychology*, 89, 800–816.

- Hao, Z., Seibert, S. E., & Hills, G. E. (2005). The mediating role of self-efficacy in the development of entrepreneurial intentions. *Journal of Applied Psychology*, 90, 1265–1272.
- Heggestad, E. D., & Kanfer, R., (2005). The predictive validity of self-efficacy in training performance: Little more than past performance. *Journal of Experimental Psychology: Applied*, 11, 84–97.
- Hendricks, J. W., & Payne, S. C. (2007). Beyond the Big Five: Leader goal orientation as a predictor of leadership effectiveness. *Human Performance*, 20, 317–344.
- Herold, D. M., Fedor, D. B., & Caldwell, S. D. (2007). Beyond change management: A multilevel investigation of contextual personal influences on employees' commitment to change. *Journal of Applied Psychology*, 92, 942–951.
- Higgins, E. T. (1997). Beyond pleasure and pain. *American Psychologist*, 52, 1280–1300.
- Holland, R. W., Hendriks, M., & Aarts, H. (2005). Smells like clean spirit: Nonconscious effects of scent on cognition and behavior. *Psychological Science*, 16, 689–693.
- Hsee, C. K., & Abelson, R. P. (1991). The velocity relation: Satisfaction as a function of the first derivative of outcome over time. *Journal of Personality and Social Psychology*, 60, 341–347.
- Hulleman, C. S., Schrager, S. M., Bodmann, S. M., & Harackiewicz, J. M. (2010). A meta-analytic review of achievement goal measures: Different labels for the same constructs or different constructs with similar labels? *Psychological Bulletin*, 136 (3), 422–449.
- Hyland, M. E. (1988). Motivational control theory: An integrative framework. *Journal of Personality and Social Psychology*, 55, 642–651.
- Iaffaldano, M. T., & Muchinsky, P. M. (1985). Job satisfaction and job performance: A meta-analysis. *Psychological Bulletin*, 97, 251–273.
- Ilgen, D. R., & Hulin, C. L. (2000). Computational modeling of behavior in organizations: The third scientific discipline. Washington, DC: American Psychological Association.
- Ilies, R., & Judge, T. A. (2005). Goal regulation across time: The effect of feedback and affect. *Journal of Applied Psychology*, 90, 453–467.
- Jagacinski, C. M., Kumar, S., Boe, J. L., Lam, H., & Miller, S. A. (2010). Changes in achievement goals and competence perceptions across the college semester. *Motivation and Emotion*, 34, 191–204.
- Jagacinski, R. J., & Flach, J. M., (2003). Control theory for humans: Quantitative approaches to modeling performance. Mahwah, NJ: Erlbaum.
- Jenkins, G. D., Mitra, A., Gupta, N., & Shaw, J. D. (1998). Are financial incentives related to performance? A meta-analytic review of empirical research. *Journal of Applied Psychology*, 83, 777–787.
- Johnson, H. M., & Spector, P. E. (2007). Service with a smile: Do emotional intelligence, gender, and autonomy moderate the emotional labor process? *Journal of Occupational Health Psychology*, 12, 319–333.
- Johnson, R. E., Chang, C.-H. D., & Lord, R. G. (2006). Moving from cognition to behavior: What the research says. *Psychological Bulletin*, 132, 381–415.
- Johnson, R. E., Rosen, C. C., & Levy, P. E. (2008). Getting to the core of core self-evaluations: A review and recommendations. *Journal of Organizational Behavior*, 29, 391–413.

- Judge, T. A., Bono, J. E., Erez, A., & Locke, E. A. (2005). Core self-evaluations and job and life satisfaction: The role of self-concordance and goal attainment. *Journal of Applied Psychology*, 90, 257–268.
- Judge, T. A., Hurst, C., & Simon, L. N. (2009). Does it pay to be smart, attractive, or confident (or all three)?: Relationships among general mental ability, physical attractiveness, core self-evaluations, and income. *Journal of Applied Psychology*, 94, 742–755.
- Judge, T. A., Jackson, C., Shaw, J. C., Scott, B. A., & Rich, B. L. (2007). Self-efficacy and work-related performance: The integral role of individual differences. *Journal of Applied Psychology*, 92, 107–127.
- Judge, T. A., Martocchio, J. J., & Thoresen, C. J. (1997). Five-factor model of personality and employee absence. *Journal of Applied Psychology*, 82, 745–755.
- Judge, T. A., Thoresen, C. J., Bono, J. E., & Patton, G. K. (2001). The job satisfaction–job performance relationship: A qualitative and quantitative review. *Psychological Bulletin*, 127, 376–407.
- Kacmar, K. M., Collins, B. J., Harris, K. J., & Judge, T. A. (2009). Core self-evaluations and job performance: The role of the perceived work environment. *Journal of Applied Psychology*, 94, 1572–1580.
- Kahneman, D., & Tversky, A. (1984). Choices, values and frames. *American Psychologist* 39, 341–350.
- Kammeyer-Mueller, J. D., Judge, T. A., & Scott, B. A. (2009). The role of core self-evaluations in the coping process: Testing an integrative model. *Journal of Applied Psychology*, 94, 177–195.
- Kanfer, R. (1990). Motivation theory and industrial/organizational psychology. In M. D. Dunnette & L. Hough (Eds.), *Handbook of industrial and organizational psychology: Vol. 1. Theory in industrial and organizational psychology* (pp. 75–170). Palo Alto, CA: Consulting Psychologists Press.
- Kanfer, R., & Ackerman, P. L. (1989). Motivation and cognitive abilities: An integrative/aptitude-treatment interaction approach to skill acquisition. *Journal of Applied Psychology*, 74, 657–690.
- Kanfer, R., Chen, G., & Pritchard, R. D. (2008). The three C's of work motivation: Context, content, and change. In R. Kanfer, G. Chen, & R. D. Pritchard (Eds.), *Work motivation: Past, present, and future* (pp. 1–16). New York, NY: Psychology Press.
- Kaplan, S., Bradley, J. C., Luchman, J. N., & Haynes, D. (2009). On the role of positive and negative affectivity in performance: A meta-analytic investigation. *Journal of Applied Psychology*, 94, 162–176.
- Kark, R., & Van Dijk, D. (2007). Motivation to lead motivation to follow: The role of the self-regulatory focus in leadership processes. *Academy of Management Review*, 32, 500–528.
- Kernan, M. C., & Lord, R. G. (1990). Effects of valence, expectancies, and goal-performance discrepancies in single and multiple goal environments. *Journal of Applied Psychology*, 75, 194–203.
- Klein, H. J. (1991). Further evidence on the relationship between goal setting and expectancy theories. *Organizational Behavior and Human Decision Processes*, 49, 230–257.
- Klein, H. J., Austin, J. T., & Cooper, J. T. (2008). Goal choice and decision processes. In R. Kanfer, G. Chen, & R. Pritchard (Eds.), *Work motivation: Past, present, and future* (pp. 101–150). Routledge/Taylor and Francis Group.
- Klein, H. J., Wesson, M. J., Hollenbeck, J. R., & Alge, B. J. (1999). Goal commitment and the goal-setting process: Conceptual clarification and empirical synthesis. *Journal of Applied Psychology*, 84, 885–896.
- Kluger, A. N., & DeNisi, A. (1996). The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological Bulletin*, 119, 254–284.

- Kluger, A. N., & Tikochinsky, J. (2001). The error of accepting the “theoretical” null hypothesis: The rise, fall, and resurrection of common sense hypotheses in psychology. *Psychological Bulletin*, 127, 408–423.
- Kozlowski, S. W. J., & Bell, B. S. (2006). Disentangling achievement orientation and goal setting: Effects on self-regulatory processes. *Journal of Applied Psychology*, 91, 900–916.
- Kozlowski, S. W., Gully, S. M., Brown, K. G., Salas, E., Smith, E. M., & Nason, E. R. (2001). Effects of training goals and goal orientation traits on multidimensional training outcomes and performance adaptability. *Organizational Behavior & Human Decision Processes*, 85, 1–31.
- Kruger, J., & Evans, M. (2004). If you don't want to be late, enumerate: Unpacking reduces the planning fallacy. *Journal of Experimental Social Psychology*, 40, 586–598.
- Kuhl, J. (1994). A theory of action and state orientation. In J. Kuhl & J. Beckmann, J. (Eds.), *Volition and personality: Action versus state orientation* (pp. 9–46). Seattle, WA: Hogrefe & Huber.
- Kuncel, N. R., & Klieger, D. M. (2007). Application patterns when applicants know the odds: Implications for selection research and practice. *Journal of Applied Psychology*, 92, 586–593.
- Larrick, R. P., Heath, C., & Wu, G. (2009). Goal-induced risk taking in negotiation and decision making. *Social Cognition*, 27, 342–364.
- Latham, G. P., & Seijts, G. H. (1999). The effects of proximal and distal goals on performance on a moderately complex task. *Journal of Organizational Behavior*, 20, 421–429.
- Lawrence, J. W., Carver, C. S., & Scheier, M. F. (2002). Velocity toward goal attainment in immediate experience as a determinant of affect. *Journal of Applied Social Psychology*, 32, 788–802.
- Lazarus, R. (1999). *Stress and emotion*. New York, NY: Springer.
- Lazarus, R. S. (1975). The self-regulation of emotion. In L. Levi (Ed.), *Emotions: Their parameters and measurement*. New York, NY: Raven Press.
- Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. New York, NY: Springer.
- Leroy, S. (2009). Why is it so hard to do my work? The challenge of attention residue when switching between work tasks. *Organizational Behavior and Human Decision Processes*, 109, 168–181.
- Leroy, S. (2010). Being present: The challenge of anticipated time pressure at work. In J. W. Beck & A. M. Schmidt (Chairs), *Any which way you can: Resource allocation among competing demands*. Symposium presented at the annual meeting of the Academy of Management, Montreal, Canada.
- Lewin, K. (1935). *A dynamic theory of personality*. New York, NY: McGraw-Hill.
- Lewin, K., Dembo, T., Festinger, L., & Sears, P. (1944). Level of aspiration. In J. M. Hunt (Ed.), *Personality and the behavior disorders* (pp. 333–378). Oxford, England: Ronald Press.
- Locke, E. A., & Latham, G. P. (1990). *A theory of goal setting and task performance*. Englewood Cliffs, NJ: Prentice Hall.
- Locke, E. A., & Latham, G. P. (2002). Building a practically useful theory of goal setting and task motivation: A 35 year odyssey. *American Psychologist*, 57, 705–717.
- Locke, E. A., & Latham, G. P. (2009). Has goal setting gone wild, or have its attackers abandoned good scholarship? *Academy of Management Perspectives*, 23, 17–23.

- Locke, E. A., Shaw, K. N., Saari, L. M., & Latham, G. P. (1981). Goal setting and task performance: 1969–1980. *Psychological Bulletin*, 90, 125–152.
- Locke, E. A., Smith, K. G., Erez, M., Chah, D., & Schaffer, A. (1994). The effects of intra-individual goal conflict on performance. *Journal of Management*, 20, 67–91.
- Lord, R. G., & Brown, D. J. (2004). Leadership processes and follower self-identity. Mahwah, NJ: Erlbaum.
- Lord, R. G., Diefendorff, J. M., Schmidt, A. M., & Hall, R. G. (2010). Self-regulation at work. *Annual Review of Psychology*, 61, 548–568.
- Lord, R. G., & Kanfer, R. (2002). Emotions and organizational behavior. In R. G. Lord, R. J. Klimoski, & R. Kanfer (Eds.), *Emotions in the workplace: Understanding the structure and role of emotions in organizational behavior* (pp. 5–19). San Francisco, CA: Jossey-Bass.
- Lord, R. G., & Levy, P. E. (1994). Moving from cognition to action: A control perspective. *Applied Psychology: An International Review*, 43, 335–398.
- Louro, M. J. S., Pieters, R., & Zeelenberg, M. (2007). Dynamics of multiple goal pursuit. *Journal of Personality and Social Psychology*, 93, 174–193.
- Ludwig, T. D., & Geller, E. S. (1997). Assigned versus participative goal setting and response generalization: Managing injury control among professional pizza deliverers. *Journal of Applied Psychology*, 82, 253–261.
- Markus, H., & Wurf, E. (1987). The dynamic self-concept: A social psychological perspective. *Annual Review of Psychology*, 38, 299–337.
- Mathieu, J. E., Rapp, T. L., Maynard, M. T. & Mangos, P. M. (2010). Interactive effects of team and task shared mental models as related to air traffic controllers' collective efficacy and effectiveness. *Human Performance*, 23, 22–40.
- McAdams, D. P., & Olson, B. D. (2011). Personality development: Continuity and change over the life course. *Annual Review of Psychology*, 61, 517–542.
- McCrae, R. R., Terracciano, A., & 79 Members of the Personality Profiles of Cultures Project (2005). Personality profiles of cultures: Aggregate personality traits. *Journal of Personality and Social Psychology*, 89, 407–425.
- McCrea, S. M., Liberman, N., Trope, Y., & Sherman, S. J. (2008). Construal level and procrastination. *Psychological Science*, 19, 1308–1314.
- Mento, A. J., Steele, R. P., & Karren, R. J. (1987). A meta-analytic study of the effects of goal setting on task performance: 1966–1984. *Organizational Behavior and Human Decision Processes*, 39, 52–83.
- Mesmer-Magnus, J. R., & Viswesvaran, C. (2005). Convergence between measures of work-to-family and family-to-work conflict: A meta-analytic examination. *Journal of Vocational Behavior*, 67, 215–232.
- Miner, A. G., Glomb, T. M. (2010). State mood, task performance, and behavior at work: A within-persons approach. *Organizational Behavior and Human Decision Processes*, 112, 43–57.
- Mischel, W. (2004). Toward an integrative science of the person (Prefatory chapter). *Annual Review of Psychology*, 55, 1–22.
- Mitchell, T. R., & Daniels, D. (2002). Motivation. In W. Borman, D. Ilgen, & R. Klimoski (Eds.), *Comprehensive handbook of psychology* (Vol. 12, pp. 225–254). New York, NY: Wiley.
- Mitchell, T. R., Harman, W. S., Lee, T. W. & Lee, D. Y. (2008). Self-regulation and multiple deadline goals. In R. Kanfer, G. Chen, & R. D. Pritchard (Eds.), *Work motivation: Past, present, and future*. Mahwah, NJ: Erlbaum.

- Mitchell, T. R., & James, L. R. (2001). Building better theory: Time and the specification of when things happen. *Academy of Management Review*, 26, 530-547.
- Moon, S. M., & Lord, R. G. (2006). Individual differences in automatic and controlled regulation of emotion and task performance. *Human Performance*, 19, 327-356.
- Moritz, S. E., Feltz, D. L., Fahrback, K. R., & Mack, D. E. (2000). The relation of self-efficacy measures to sport performance: a meta-analytic review. *Research Quarterly for Exercise and Sport*, 71, 280.
- Mount, M., Ilies, R., & Johnson, E. (2006). Relationship of personality traits and counterproductive work behaviors: The mediating effects of job satisfaction. *Personnel Psychology*, 59, 591-622.
- Multon, K. D., Brown, S. D., & Lent, R. W. (1991). Relation of self-efficacy beliefs to academic outcomes: A meta-analytic investigation. *Journal of Counseling Psychology*, 38, 30-38.
- Nahrgang, J. D., Morgeson, F. P., & Hofmann, D. A. (2011). Safety at work: A meta-analytic investigation of the link between job demands, job resources, burnout, engagement, and safety outcomes. *Journal of Applied Psychology*, 96, 71-94.
- Neubert, M. J., Kacmar, M., Carlson, D. S., Chonko, L. B., & Roberts, J. A. (2008). Regulatory focus as a mediator of the influence of initiating structure and servant leadership on employee behavior. *Journal of Applied Psychology*, 93, 1220-1233.
- Ng, K. Y., Ang, S., & Chan, K. Y. (2008). Personality and leader effectiveness: A moderated mediation model of leadership self-efficacy, job demands and job autonomy. *Journal of Applied Psychology*, 93, 733-743.
- Northcraft, G., Schmidt, A. M., & Ashford, S. J. (2011). Feedback and the rationing of time and effort among competing tasks. *Journal of Applied Psychology*, 96 (5), 1076-1086.
- Ones, D. S., & Dilchert, S., Viswesvaran, C., & Judge, T. A. (2007). In support of personality assessment in organizational settings. *Personnel Psychology*, 60, 995-1027.
- Ordóñez, L. D., Schweitzer, M. E., Galinsky, A. D., & Bazerman, M. H. (2009). Goals gone wild: The systematic side effects of overprescribing goal setting. *Academy of Management*, 23, 6-16.
- Orehek, E., Bessarabova, E., Chen, X., & Kruglanski, A. W. (2011). Positive affect as informational feedback in goal pursuit. *Motivation and Emotion*, 35, 44-51.
- Panksepp, J. (1998). *Affective neuroscience: The foundations of human and animal emotions*. Oxford, England: Oxford University Press.
- Panksepp, J. (2005). On the embodied neural nature of core emotional affects. *Journal of Consciousness Studies*, 12, 158-184.
- Panksepp, J. (2008). The affective brain and core consciousness: How does neural activity generate emotional feelings? In M. Lewis, J. M. Haviland, & L. F. Barrett (Eds.), *Handbook of emotions* (pp. 46-67). New York, NY: Guilford Press.
- Panksepp J. (2010, February 15). Jaak Panksepp's affective neuroscience. Retrieved from <http://integral-options.blogspot.com/2010/02/jaak-panksepps-affective-neuroscience.html>
- Park, G., Schmidt, A. M., Scheu, C. R., & DeShon, R. P. (2007). Process model of feedback seeking. *Human Performance*, 20, 119-145.
- Payne, S. C., Youngcourt, S. S., & Beaubien, J. M. (2007). A meta-analytic examination of the goal orientation nomological net. *Journal of Applied Psychology*, 92, 128-150.
- Perry, S. J., Hunter, E. M., Witt, L. A., & Harris, K. J. (2010). $P = f(\text{Conscientiousness} \times \text{Ability})$: Examining the facets of conscientiousness. *Human Performance*, 23, 343-360.

- Peterson, S. J., & Luthans, F. (2006). The impact of financial and non-financial incentives on business unit outcomes over time. *Journal of Applied Psychology*, 91, 156–165.
- Ployhart, R. E. (2008). Work motivation methods, measures, and assessment strategies. In R. Kanfer, G. Chen, & R. Pritchard (Eds.), *Work motivation: Past, present, and future* (pp. 17–61). New York, NY: Routledge.
- Porter, C. O. L. H. (2005). Goal orientation: Effects on backing up behavior, performance, efficacy, and commitment in teams. *Journal of Applied Psychology*, 90, 811–811.
- Porter, C. O. L. H., Webb, J. W., & Gogus, C. I. (2010). When goal orientations collide: Effects of learning and performance orientation on team adaptability in response to workload imbalance. *Journal of Applied Psychology*, 95, 935–943.
- Porter, L., & Lawler, E. (1968). *Managerial attitudes and performance*. Homewood, IL: Irwin Dorsey.
- Powers, W. T. (1973). *Behavior: The control of perception*. Chicago, IL: Aldine.
- Pritchard, R. D., & Ashwood, E. L. (2007). *Managing motivation: A manager's guide to diagnosing and improving motivation*. New York, NY: LEA/Psychology Press.
- Pugh, S. D. (2002). Emotional regulation in individuals and dyads: Causes, costs, and consequences. In R. G. Lord, R. Klimoski, & R. Kanfer (Eds.), *Emotions in the workplace* (pp. 147–182). San Francisco, CA: Jossey-Bass.
- Reinhard, M., & Dickhäuser, O. (2009). Need for cognition, task difficulty, and the formation of performance expectancies. *Journal of Personality and Social Psychology*, 96, 1062–1076.
- Reynolds, S. J. (2006). A neurocognitive model of the ethical decision-making process: Implications for study and practice. *Journal of Applied Psychology*, 91, 737–748.
- Richard, E. M., & Diefendorff, J. M. (2011). Self-regulation during a single performance episode: Mood as information in the absence of formal feedback. *Organizational Behavior and Human Decision Processes*, 1, 99–110.
- Richard, E. M., Diefendorff, J. M., & Martin, J. H. (2006). Revisiting the within-person self-efficacy and performance relationship. *Human Performance*, 19, 67–87.
- Riketta, M. (2008). The causal relationship between job attitudes and performance: A meta-analysis of panel studies. *Journal of Applied Psychology*, 93, 472–481.
- Rogelberg, S. G., Leach, D. J., Warr, P. B., & Burnfield, J. L. (2006). “Not another meeting!” Are meeting time demands related to employee well-being? *Journal of Applied Psychology*, 1, 86–96.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55, 68–78.
- Scherbaum, C. A., & Vancouver, J. B. (2010). If we produce discrepancies, then how? Testing a computational model of positive goal revision. *Journal of Applied Social Psychology*, 40, 2201–2231.
- Schmidt, A. M., & Beck, J. W. (2011). Goal difficulty moderating self-efficacy's relationship with effort and performance. Paper to be presented at the 26th Annual Conference of the Society for Industrial and Organizational Psychology, Chicago, IL.
- Schmidt, A. M., & DeShon, R. P. (2007). What to do? The effects of discrepancies, incentives, and time on dynamic goal prioritization. *Journal of Applied Psychology*, 92, 928–941.
- Schmidt, A. M., & DeShon, R. P. (2009). Prior performance and goal progress as moderators of the relationship between self-efficacy and performance. *Human Performance*, 22, 191–203.

- Schmidt, A. M., & DeShon, R. P. (2010). The moderating effects of performance ambiguity on the relationship between self-efficacy and performance. *Journal of Applied Psychology*, 95, 572–581.
- Schmidt, A. M., Dolis, C. M., & Tolli, A. P. (2009). A matter of time: Individual differences, contextual dynamics, and goal progress effects on multiple-goal self-regulation. *Journal of Applied Psychology*, 94, 692–709.
- Schmidt, A. M., & Dolis, C. M. (2009). Something's got to give: The effects of dual-goal difficulty, goal progress, and expectancies on resource allocation. *Journal of Applied Psychology*, 94, 678–691.
- Schweitzer, M., Ordóñez, L. D., & Douma, B. (2004). The dark side of goal setting: The role of goals in motivating unethical behavior. *The Academy of Management Journal*, 47, 422–432.
- Seijts, G. H., Latham, G. P., Tasa, K., & Latham, B. W. (2004). Goal setting and goal orientation: An integration of two different yet related literatures. *Academy of Management Journal*, 47, 227–239.
- Senko, C. M., & Harackiewicz, J. M. (2005). Achievement goals, performance and task interest: Why perceived difficulty matters. *Personality and Social Psychology Bulletin*, 31, 1739–1753.
- Seo, M., & Ilies, R. (2009). The role of self-efficacy, goal, and affect in dynamic motivational self-regulation. *Organizational Behavior and Human Decision Processes*, 109, 120–133.
- Seo, M-G., Goldfarb, B., & Barrett, L. F. (2010). Affect and the framing effect within individuals across time: Risk taking in a dynamic investment game. *Academy of Management Journal*, 53, 411–431.
- Shah, J., & Higgins, E. T. (1997). Expectancy value effects: Regulatory focus as determinant of magnitude and direction. *Journal of Personality & Social Psychology*, 73, 447–458.
- Shah, J. Y., Friedman, R., & Kruglanski, A. W. (2002). Forgetting all else: On the antecedents and consequences of goal shielding. *Journal of Personality and Social Psychology*, 83, 1261–1280.
- Shah, J. Y., & Kruglanski, A. W. (2003). When opportunity knocks: Bottom-up priming of goals by means and its effects on self-regulation. *Journal of Personality and Social Psychology*, 84, 1109–1122.
- Shantz, A., & Latham, G. P. (2009). An exploratory field experiment of the effect of subconscious and conscious goals on employee performance. *Organizational Behavior and Human Decision Processes*, 109, 9–17.
- Shaw, J. D., & Gupta, N. (2007). Pay system characteristics and quit patterns of good, average, and poor performers. *Personnel Psychology*, 60, 903–928.
- Shaw, J. D., Duffy, M. K., Mitra, A., Lockhart, D., & Bowler, W. M. (2003). Reactions to merit pay increases: A longitudinal test of a signal sensitivity theory. *Journal of Applied Psychology*, 88, 538–544.
- Sheldon, K. M., Elliot, A. J., Kim, Y., & Kasser, T. (2001). What is satisfying about satisfying events? Testing 10 candidate psychological needs. *Journal of Personality and Social Psychology*, 80, 325–329.
- Smillie, L., Yeo, G., & Furnham, A., & Jackson, C. J. (2006) Benefits of all work and no play: The relationship between neuroticism and performance as a function of resource allocation. *Journal of Applied Psychology*, 91, 139–155.
- Smith, P. C., Kendall, L. M., & Hulin, C. L. (1969). The measurement of satisfaction in work and retirement. Chicago, IL: Rand McNally.
- Srivastava, A., Bartol, K. M., & Locke, E. A. (2006). Empowering leadership in management teams: Effects on knowledge sharing, efficacy, and performance. *Academy of Management Journal*, 49, 1239–1251.

- Stajkovic, A. D., Lee, D., & Nyberg, A. J. (2009). Collective efficacy, group potency, and group performance: Meta-analyses of their relationships, and test of mediation model. *Journal of Applied Psychology*, 94, 814–828.
- Stajkovic, A. D., Locke, E. A., & Blair, E. (2006). A first examination of the relationships between primed subconscious goals, assigned conscious goals, and task performance. *Journal of Applied Psychology*, 91, 1171–1180.
- Stajkovic, A. D., & Luthans, F. (1998). Self-efficacy and work related performance: A metaanalysis. *Psychological Bulletin*, 124, 240–261.
- Stajkovic, A. D., & Luthans, F. (2003). Behavioral management and task performance in organizations: Conceptual background, meta-analysis, and test of alternative models. *Personnel Psychology*, 56, 155–194.
- Steel, P. (2007). The nature of procrastination: A meta-analytic and theoretical review of quintessential self-regulatory failure. *Psychological Bulletin*, 133, 65–94.
- Steel, P. (2010). Arousal, avoidant and decisional procrastinators: Do they exist? *Personality and Individual Differences*, 48, 926–934.
- Steel, P., & König, C. J. (2006). Integrating theories of motivation. *Academy of Management Review*, 31, 889–913.
- Stewart, G. L., & Nandkeolyar, A. K. (2006). Adaptation and intraindividual variation in sales outcomes: Exploring the interactive effects of personality and environmental opportunity. *Personnel Psychology*, 59, 307–332.
- Stewart, G. L., & Nandkeolyar, A. K. (2007). Exploring how constraints created by other people influence intraindividual variation in objective performance measures. *Journal of Applied Psychology*, 92, 1149–1158.
- Tamir, M. (2005). Don't worry, be happy? Neuroticism, trait-consistent affect regulation, and performance. *Journal of Personality and Social Psychology*, 89, 449–461.
- Tolli, A. P., & Schmidt, A. M. (2008). The role of feedback, causal attributions, and self-efficacy in goal revision. *Journal of Applied Psychology*, 93, 692–701.
- Trougakos, J. P., Beal, D. J., Green, S. G., & Weiss, H. M. (2008). Making the break count: An episodic examination of recovery activities, emotional experiences, and positive affective displays. *Academy of Management Journal*, 51, 131–146.
- Tsai, W.-C., Chen, C.-C., & Liu, H.-L. (2007). Test of a model linking employee positive moods and task performance. *Journal of Applied Psychology*, 92, 1570–1583.
- Tubbs, M. E. (1986). Goal-setting: A meta-analytic examination of the empirical evidence. *Journal of Applied Psychology*, 71, 474–483.
- Turban, D. B., Stevens, C. K., & Lee, F. (2009). Effects of conscientiousness and extraversion on new labor market entrants' job search: The mediating role of meta-cognitive activities and positive emotions. *Personnel Psychology*, 63, 553–573.
- Van Hooft, E. A. J., & Noordzij, G. (2009). The effects of goal orientation on job search and reemployment: A field experiment among unemployed job seekers. *Journal of Applied Psychology*, 94, 1581–1590.
- Vancouver, J. B. (2005). The depth of history and explanation as benefit and bane for psychological control theories. *Journal of Applied Psychology*, 90, 38–52.
- Vancouver, J. B. (2008). Integrating self-regulation theories of work motivation into a dynamic process theory. *Human Resource Management Review*, 18, 1–18.
- Vancouver, J. B., & Day, D. V. (2005). Industrial and organization research on self-regulation: From constructs to applications. *Applied Psychology: An International Review*, 54, 155–185.

- Vancouver, J. B., & Kendall, L. N. (2006). When self-efficacy negatively relates to motivation and performance in a learning context. *Journal of Applied Psychology*, 91, 1146–1153.
- Vancouver, J. B., More, K. M., & Yoder, R. J. (2008). Self-efficacy and resource allocation: Support for a nonmonotonic discontinuous model. *Journal of Applied Psychology*, 93, 35–47.
- Vancouver, J. B., Putka, D. J., & Scherbaum, C. A. (2005). Testing a computational model of the goal-level effect: An example of a neglected methodology. *Organizational Research Methods*, 8 (1), 100–127.
- Vancouver, J. B., Thompson, C. M., Tischner, E. C., & Putka, D. J. (2002). Two studies examining the negative effect of self-efficacy on performance. *Journal of Applied Psychology*, 87, 506–516.
- Vancouver, J. B., Thompson, C. M., & Williams, A. A. (2001). The changing signs in the relationships among self-efficacy, personal goals, and performance. *Journal of Applied Psychology*, 86, 605–620.
- Vancouver, J. B., Weinhardt, J. M., & Schmidt, A. M. (2010). A formal, computational theory of multiple goal pursuit: Integrating goal-choice and goal-striving processes. *Journal of Applied Psychology*, 95, 985–1008.
- Van Eerde, W., & Thierry, H. (1996). Vroom's expectancy models and work-related criteria: A meta-analysis. *Journal of Applied Psychology*, 81, 575–586.
- Vogt, J., De Houwer, J., Moors, A., Van Damme, S., & Crombez, G. (2010). The automatic orienting of attention to goal-relevant stimuli. *Acta Psychologica*, 134, 61–69.
- Vroom, V. H. (1964). *Work and motivation*. New York, NY: Wiley.
- Wallace, C., & Chen, G. (2006). A multilevel integration of personality, climate, self-regulation, and performance. *Personnel Psychology*, 59, 529–557.
- Wallace, J. C., Edwards, B. D., Shull, A., & Finch, D. M. (2009). Examining the consequences in the tendency to suppress and reappraise emotions on task-related job performance. *Human Performance*, 22, 23–43.
- Wallace, C., Johnson, P., & Frazier, M. (2009). An examination of the factorial, construct, and predictive validity and utility of the regulatory focus at work scale. *Journal of Organizational Behavior*, 30, 805–831.
- Waller, M. J., Conte, J. M., Gibson, C. G., & Carpenter, M. A. (2001). The impact of individual time perception on team performance under deadline conditions. *Academy of Management Review*, 26, 586–600.
- Waller, M. J., Zellmer-Bruhn, M., & Giambatista, R. C. (2002). Watching the clock: Team pacing behaviors under deadlines. *Academy of Management Journal*, 45, 1046–1055.
- Walumbwa, F. O., Avolio, B. J., & Zhu, W. (2008). How transformational leadership weaves its influence on individual job performance: The role of identification and efficacy beliefs. *Personnel Psychology*, 61 (4), 793–825.
- Wanberg, C., Zhu, J., & Van Hooft, E. A. J. (2010). The job search grind: Perceived progress, self-reactions, and self-regulation of search effort. *Academy of Management Journal*, 53, 788–807.
- Wanberg, C. R., Glomb, T. M., Song, Z., & Sorenson, S. (2005). Job-search persistence during unemployment: a 10-wave longitudinal study. *Journal of Applied Psychology*, 90, 411–430.
- Wang, M., & Takeuchi, R. (2007). The role of goal orientation during expatriation: A cross-sectional and longitudinal investigation. *Journal of Applied Psychology*, 92, 1437–1445.
- Wegner, D. M. (1994). Ironic processes of mental control. *Psychological Review*, 101, 34–52.

- Weiss, H. M. (2002). Conceptual and empirical foundations for the study of affect at work. In R. G. Lord, R. J. Klimoski, & R. Kanfer (Eds.), *Emotions in the workplace: Understanding the structure and role of emotions in organizational behavior* (pp. 20–63). San Francisco, CA: Jossey-Bass.
- Weiss, H. M., & Beal, D. J. (2005). Reflections on affective events theory. In N. M. Askanasy, W. Zerbe, & C. E. J. Hartel (Eds.), *Research on emotion in organizations: The effect of affect in organizational settings* (Vol. 1, pp. 1–21). Oxford, England: Elsevier.
- Weiss, H. M., & Cropanzano, R. (1996). Affective events theory: A theoretical discussion of the structure, causes and consequences of affective experiences at work. *Research in Organizational Behavior*, 18, 1–74.
- Weiss, H. M., & Rupp, D. E. (2011). Experiencing work: An essay on a person-centric work psychology. *Industrial and Organizational Psychology*, 4, 83–97.
- Weyman, A. K., & Clarke, D. D. (2003). Investigating the influence of organizational role on perceptions of risk in deep coal mines. *Journal of Applied Psychology*, 88, 404–412.
- Wharton, A. S. (2009). The sociology of emotional labor. *Annual Review of Sociology*, 35, 147–165.
- Williams, K. J, Donovan, J. J, & Dodge, T. L. (2000). Self-regulation of performance: Goal establishment and goal revision processes in athletes. *Human Performance*, 13, 159–180.
- Wood, R., Mento, A., & Locke, E. (1987). Task complexity as a moderator of goal effects. *Journal of Applied Psychology*, 72, 416–425.
- Wood, R. E., & Bandura, A. (1989). Impact of conceptions of ability on self-regulatory mechanisms and complex decision making. *Journal of Personality and Social Psychology*, 56, 407–415.
- Yamagata, S., Suzuki, A., Ando, J., Ono, Y., Kijima, N., Yoshimura, K.,...Spinath, F. M. (2006). Is the genetic structure of human personality universal? A cross-cultural twin study from North America, Europe, and Asia. *Journal of Personality and Social Psychology*, 90, 987–998.
- Yeo, G., & Neal, A. (2006). An examination of the dynamic relationship between self-efficacy and performance across levels of analysis and levels of specificity. *Journal of Applied Psychology*, 91, 1088–1101.
- Yeo, G., & Neal, A. (2008). Subjective cognitive effort: A model of states, traits, and time. *Journal of Applied Psychology*, 93, 617–631.
- Yeo, G., Loft, S., Xiao, T., & Kiewitz, C. (2009). Goal orientations and performance: differential relationships across levels of analysis and as a function of task demands. *Journal of Applied Psychology*, 94, 710–726.
- Zoogah, D. B. (2010). Why should I be left behind? Employees' perceived relative deprivation and participation in development activities. *Journal of Applied Psychology*, 95, 159–173.